Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Management Science and Information Systems Studies

Final Year Project Report



Female Entrepreneurship in Tanzania: Developing a Crowdfunding Solution for KPesa

Ciaran Mc Kay

April 2023

TRINITY COLLEGE DUBLIN
Management Science and Information Systems Studies
Project Report

KPESA
Female Entrepreneurship in Tanzania: Developing a Crowdfunding Solution

April 2023

Prepared by: Ciaran McKay ***Supervisor: Paula Roberts***

Declaration

I declare that the work described in this dissertation has been carried out in full compliance with the ethical requirements of the School of Computer Science and Statistics.

I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at: http://www.tcd.ie/calendar

I have also completed the Online Tutorial on avoiding plagiarism 'Ready, Steady, Write', located at http://tcd-ie.libguides.com/plagiarism/ready-steady-write

I declare that the report being submitted represents my own work and has not been taken from the work of others save where appropriately referenced in the body of the assignment.

Signed: _____

Ciaran McKay 14/04/2023

ABSTRACT

This project investigated the feasibility of developing a crowdfunding web app designed for female entrepreneurs in Tanzania. The project scope spanned research, design, development, and testing of a web app for the client KPesa. The research phase considered the motivations and challenges female entrepreneurs face in Africa and the different funding models that could be used to support them. Applying the research, the web app was developed with a community-first approach. The features of the web app include chat feature and a campaign section where entrepreneurs seek investment. The web app has been developed using HTML, React.js, Tailwind CSS, Next.js and Firebase. The result is an intuitive, mobile-first platform that has low bandwidth requirements.

PREFACE

This project was completed on behalf of KPesa. KPesa is a social enterprise that aims to leverage the power of crowdfunding to empower female entrepreneurship in Eastern Africa, specifically Tanzania. The founder, Leyla Faru, is a native of Tanzania living in Dublin. She is a chartered accountant with experience working for KPMG in Dublin. Leyla observed the need for microfinancing for female entrepreneurs in Tanzania following a trip home in 2019 and began exploring the possibility of launching her own crowdfunding platform. Consequently, Leyla tasked me to explore the feasibility of developing a crowdfunding web app to help her establish the KPesa community. The project had two primary phases; a research and a development phase and has successfully met the terms of reference agreed upon with the client.

The research yielded an observable demand for financing in the Tanzanian market. In fact, most significant challenge for female entrepreneurs related to access to capital at the early stages of female enterprise development in Tanzania. The investigation also underscored the need for community. As such, a chat function was recommended. In addition, the analysis also revealed challenges relating to regulatory requirements and business models, which KPesa must solve before advancing with the web app.

The web app has two main features. The first feature is a community forum, and the second is a campaign section. These features were informed by the research investigating the drivers of female entrepreneurship in Africa.

If the user is an entrepreneur, they can log in and use the chat features. Otherwise, they can choose to build and launch a new campaign. Once the campaign is made, it will appear in a campaign gallery where investors can view and search for specific campaigns. Should an investor wish to invest, they can pledge a defined sum of money through KPesa's portal. The app will primarily be used on mobile devices in Tanzania, so an emphasis was placed on building an intuitive, and lightweight app.

There were many challenges faced throughout the duration of the project. One such challenge was defining the web app's requirements and communicating with the client as they were based in Tanzania for an extended period, with limited access to reliable internet for meetings. This led to a delay in the initial phase of the project. The empirical evidence experienced through our communications challenges helped me better understand the difficulties faced by the users I was developing the app for. The lost time was effectively recovered through strong, more frequent communication with the client towards the later stages of the project.

I want to thank Paula Roberts for her guidance, assistance, and reassurance throughout the project. I would also like to thank KPesa and Leyla for their cooperation. I really enjoyed working on such an impactful project, and it would not have been possible without either party.

KPESA

FEMALE ENTREPRENEURSHIP IN TANZANIA: DEVELOPING A CROWDFUNDING SOLUTION

14th April 2023

TABLE OF CONTENTS

NO.	SECTION	PAGE
1.	INTRODUCTION & SUMMARY	**1**
1.1	The Client Company	1
1.2	Project Background	1
1.3	Terms of Reference	2
1.4	Summary of Chapters	2
2.	FEASIBILITY INVESTIGATION OF CROWDFUNDING IN TANZANIA	**2**
2.1	Exploration of the Factors Influencing Female Entrepreneurship in Tanzania	3
2.2	Considerations for KPesa's Crowdfunding Platform	5
3.	WEB APP OVERVIEW	**8**
3.1	Web App Objectives	8
3.2	Technical Environment	8
3.3	Web App Description	10
3.4	Web App Overview Diagram	11
4.	BUILDING THE WEB APP	**13**
4.1	Defining the Web App Requirements	13
4.2	Choosing Suitable Technology	13
4.3	Designing the Web App	15
4.4	Developing the Web app	17
4.5	Testing the Web app	22
4.6	Challenges Encountered	23
5.	CONCLUSION AND RECOMMENDATIONS	**24**
5.1	Conclusion	24
5.2	Recommendations	24

APPENDICES

NO.	SECTION	PAGE
A.	ORIGINAL PROJECT OUTLINE	A.1
B.	INTERIM REPORT	B.1
C.	DESIGN DOCUMENTATION	C.1
C.1	Entity Relationship Diagram and Relational Schema	C.1
C.2	Data Flow Diagrams	C.2
C.3	Mock-ups	C.2
C.4	Software Development Methodology	C.2
C.5	Coding Conventions Used	C.3
D.	SAMPLE OF THE APPLICATION SOURCE CODE & INPUT SCREENS	D.1
D.1	Sample of the Application Source Code	D.1
D.2	Sample Input Screens & Video Demonstration	D.1
E.	TESTING	E.1
E.1	Test Documentation Tables	E.1
E.2	Vercel Deployment Analytics	E.2
F.	ADDITIONAL MATERIAL PRODUCED FOR THE CLIENT	F.1
F.1	Firebase Administrator Guide	F.1
F.2	Exploration of Payment Gateways	F.1
F.3	Crowdfunding Business Models	F.2
F.4	Survey of Female Entrepreneurs in Tanzania	F.3
G.	NEXT STEPS AND CLIENT FEEDBACK	G.1
G.1	Next Steps	G.1
G.2.	Client Feedback	G.1
H.	REFERENCES	H.1

TABLE OF FIGURES

NO.	FIGURE	PAGE
FIGURE 2.2.1:	SOLUTIONS TO CHALLENGES OF FEMALE ENTREPRENEURS	5
FIGURE 2.2.2:	COMPARISON OF CROWDFUNDING BUSINESS MODELS	6
FIGURE 2.2.3:	OVERVIEW OF LOAN-BASED CROWDFUNDING MODEL	6
FIGURE 2.3.4:	SUMMARY OF TANZANIA'S CROWDFUNDING REGULATION	7
FIGURE 3.2.1:	COMPARISON OF CLIENT-SIDE RENDERING VS PRE-RENDERING	9
FIGURE 3.2.2:	SNIPPET OF CODE DEMONSTRATING TAILWIND CSS	10
FIGURE 3.3.1:	MOBILE USER'S EXPERIENCE OF THE CHAT FEATURE	10
FIGURE 3.3.2:	MOBILE USER'S EXPERIENCE OF THE CAMPAIGN FEATURE	11
FIGURE 3.4.1:	ENTITY INTERACTION DIAGRAM	12
FIGURE 3.4.2:	SITEMAP	12
FIGURE 4.3.1:	SAMPLE CODE USED FOR AUTHENTICATION	16
FIGURE 4.3.2:	SUBCOLLECTION DEMONSTRATION	16
FIGURE 4.3.3:	ACTIVITY FLOW DIAGRAM	17
FIGURE 4.4.1	MULTI-STAGE PAGE STEPPER	19
FIGURE 4.4.2	IMAGE UPLOAD & RETRIEVAL HANDLER	20

1. INTRODUCTION & SUMMARY

In this chapter, the client is introduced, and the project context is provided. The terms of reference which were agreed upon, are also provided. Additionally, a brief overview of the upcoming chapters is included.

1.1 The Client Company

KPesa, the client, is a social enterprise investigating the power of crowdfunding to bridge the financing gap for women led micro, small and medium-sized businesses (MSMEs). Africa is the only continent in the world where female entrepreneurs outnumber men, yet many remain unbanked and cannot access loans through traditional financial institutions (Dushime, 2022). KPesa aims to overcome this obstacle by building a solution that will promote economic growth by raising finance from the public to provide low-cost finance to female led MSMEs, helping female entrepreneurs grow their businesses from the informal sector to the formal sector.

1.2 Project Background

This project built an online community for KPesa, where they help entrepreneurs raise funding through formal channels. The client required a web app where users can log on using their mobile phones. Once logged in, they can access a community forum and a crowdfunding page where campaigns are launched and viewed, and pledges can be made.

There are two significant challenges for female entrepreneurs in Africa. The first challenge is the motivations behind entrepreneurship, and the second challenge is the availability of internet access.

Entrepreneurs in economically disadvantaged regions have different motivations than entrepreneurs in developed markets, who are primarily driven by opportunity. In contrast, entrepreneurs in developing markets are driven by necessity (Giacomin et al., 2011). Opportunity entrepreneurs prefer to invest in longer-term future opportunities, while survival instincts drive need entrepreneurs. By building a community where education and mentorship are core foundations, funding solutions like KPesa can help users move from need entrepreneurship to opportunity entrepreneurship.

Secondly, with only 22% of the population having access to the internet, Africa is the continent with the lowest level of connectivity. In contrast, Europe has a significantly higher level of connectivity, with over 80% of the population having access to the internet. KPesa, launching its services in Tanzania, targets a population where nearly 90% have a mobile cellular connection, but only 2% have access to fixed broadband (World Bank, 2021). As a result, internet access is limited to just 37% of the population, with most web pages being accessed via mobile phones, accounting for 81% (Mwakideu, 2021; Kemp, 2022).
Even those with access to the internet have limited data; therefore, web pages serving the region must be mobile-first and lightweight.

1.3 Terms of Reference

The agreed terms of reference with the client were as follows:

1. Explore the factors influencing female entrepreneurship in Tanzania and how KPesa can implement features to address these needs.
2. Examine the different crowdfunding business models available and offer guidance on appropriate solutions.
3. Build a full-stack web app that will be used for KPesa's crowdfunding platform.
4. Ensure the web app remains lightweight by avoiding superfluous features that may add bloat.
5. Build a chat feature within the web app where members can communicate in a forum-based community.
6. Build a design prototype of a fundraising platform where entrepreneurs can post campaigns and access funding through investors.

1.4 Summary of Chapters

Chapter 2 explores the feasibility of a crowdfunding platform in Tanzania. It examines the motivations behind female entrepreneurship and the challenges faced by female entrepreneurs. Finally, the regulatory context of Tanzania is summarised, and recommendations are offered for the business model, features, and regulatory requirements.

Chapter 3 provides a breakdown of the web app. It outlines the objectives of the web app and the technical environment; the web app overview is provided along with a web app overview diagram and sitemap.

Chapter 4 describes the work completed throughout the development phase of the project. The chapter defines web app requirements and the work completed designing, building, and testing the final product.

Chapter 5 summarises the conclusions and recommendations drawn from the project. The summary breaks down the completed sections and summarises the project, while the recommendation offers solutions to improve the web app's quality.

Appendices provide additional context to the report. It includes sections such as the original project outline, the interim report, design documentation, input screens, test documentation, and research.

2. INVESTIGATING CROWDFUNDING FOR FEMALE ENTREPRENEURS IN TANZANIA

This section explores the feasibility of launching a crowdfunding platform in Tanzania. The research conducted in this section drove the development of the web app and the recommendations made. The drivers and challenges of female entrepreneurship, crowdfunding business models and the regulatory environment in Tanzania are investigated. Based on the research, recommendations have been made to KPesa on launching their web-app in Tanzania.

2.1 Exploration of the Factors Influencing Female Entrepreneurship in Tanzania

Female Entrepreneurship

To build a high-quality solution for female entrepreneurs, it is an imperative to be informed by the drivers and challenges of female entrepreneurship. Amon et al. (2021) state that a thorough understanding of women's motives for starting their own businesses is necessary to effectively develop programmes aimed at female entrepreneurs. Therefore, the project's first phase focused on gaining an appreciation for factors driving female entrepreneurship.

McAdam (2022) notes that men are twice as likely as women to be in the process of launching new ventures but the literature available on female entrepreneurship has increased significantly in recent years. However, the focus remains primarily on females' challenges and barriers rather than female entrepreneurs' motivations.

The discourse around motivations suggests many factors. In some instances, flexibility, job satisfaction and quality of life have been cited as significant in the entrepreneurial decisions made by women (McAdam, 2022). Specifically, the literature refers to the concept of push and pull factors. Pull factors attract people to start business activities, while push factors force individuals into self-employment (Amon et al., 2021). Pull factors include autonomy, independence, social status, and greater personal control, while push factors typically relate to employee dissatisfaction, insufficient salary, and necessity.

It has been widely cited that women in less developed economies are motivated by necessity. In contrast, women in developed economies are more likely to start a business out of opportunity (McAdam, 2022). The idea of necessity entrepreneurship is supported by an extensive study completed in Tanzania which found that the motivations for starting a business are primarily push factors (Isaga, 2018). Four of the five most influential motivations were push factors, which encompassed the need to earn a living, the inability to depend on their partner's income, the need to augment their own income, and the inability to secure employment. Some less influential factors include being their own boss, achieving a higher position in society, continuing a family tradition and gaining recognition for their accomplishments (Isaga, 2018). Further detail on the motivating factors of female entrepreneurship can be found in Appendix F.4.

Challenges Facing Female Entrepreneurs

The challenges facing female entrepreneurs are well-cited in academic literature. Female entrepreneurs are frequently limited in the access they have to financial capital. There are many reasons for this. One major contributing factor is that female entrepreneurs in developing nations typically have lower levels of education than their male counterparts (Amon et al., 2021). Isaga (2018) found in their study in Tanzania that just 1.3% of female entrepreneurs held a bachelor's degree, while 58% had either a primary-level education or no education. Akinbami and Aransiola (2015) support this claim saying that female entrepreneurs' low level of education is visible in rural areas, with microentrepreneurs often having no or only some primary-level education. Mersha and Sriram (2019) have highlighted that female entrepreneurs may experience challenges related to a perceived lack of education and experience, which can negatively affect their confidence in succeeding. Additionally, as observed in previous studies, these entrepreneurs may exhibit a higher fear of failure and view success as being outside of their control.

In addition, educational barriers make it difficult for females to process the paperwork necessary for bank loans. High-interest rates, personal guarantee requirements, and gender discrimination make it very difficult for female entrepreneurs to access the capital (Akinbami and Aransiola, 2015). Tanzanian entrepreneurs cited the three biggest challenges were all related to the ability to access capital. A survey participant claimed that she raises finance from informal institutions as the bank interest rates are "30%-50% per month", making it impossible for her to expand her business. To verify this statement, I accessed a loan calculator on CBRD, the largest bank in Tanzania. The smallest business loan available was €20,000 (50,000,000 TZS), and the cheapest interest rate was 20%. My estimated monthly repayment was €520 (1,324,694 TZS); *annual repayments would be more than six times the GDP per capita of Tanzania*, excluding many microentrepreneurs (CRDB Bank, 2023).

Further, it has been reported in studies of African entrepreneurs that females' returns from loans, measured by the rate they increase revenue or profitability, are lower than their male counterparts (Bardasi et al., 2011; Witbooi and Ukpere, 2011). The authors attributed this disparity to heightened risk aversion exhibited by female entrepreneurs which aligns with the findings of earlier literature. These observations highlight the need for a comprehensive approach to crowdfunding that can effectively promote female opportunity entrepreneurship.

The Power of Community

The power of community in entrepreneurship cannot be understated in Eastern Africa. The literature has emphasised the importance of networks and role models for improving entrepreneurial performance (Brüderl and Preisendörfer, 1998). An apt example of the power of a network to support an idea is detailed in a study conducted on the use of WhatsApp as a platform for crowdfunding in Tanzania. Ngowi (2021) observed 18 WhatsApp group chats and found that €30,600 (77,750,606 TZS) was raised for various purposes. The author noted that many contributors were from the beneficiaries' social networks. Regulatory oversight issues are less relevant in instances like this where there is a high level of accountability within a

community setting. Community settings such as WhatsApp or churches have been seen as key drivers of the idea-fruition process for female entrepreneurs in Africa (Halberstadt and Spiegler, 2018; Amon et al., 2021). The challenge for female entrepreneurs is that due to traditional responsibilities and poor infrastructural support specific to female entrepreneurs, they typically cannot dedicate enough time to business-related activities such as networking and mentoring, which hampers their performance (Panda, 2018).

2.2 Considerations for KPesa's Crowdfunding Platform

When considering the focus of KPesa's platform, there were three important considerations: addressing the challenges of female entrepreneurship, choosing an appropriate crowdfunding model, and addressing regulatory requirements. This section explores the evaluation made with respect to KPesa and the considerations made to guide the development of the web app.

Female Entrepreneurship

In Section 2.1, the challenges faced by female entrepreneurs were addressed. The key challenges faced were raising capital, their attitude to entrepreneurship, inability to access a supportive community and societal issues. The solutions to address these challenges are summarised in FIGURE 2.2.1.

Challenge	Description	Proposed Solution
Access to capital	Female entrepreneurs frequently cite access to capital as a constraint	KPesa is building a crowdfunding platform specifically for female entrepreneurs
Attitude to entrepreneurship	Many view entrepreneurship as a mode of survival rather than opportunity	KPesa should focus on running educational campaigns focused on highlighting the opportunities that come with entrepreneurship
Access to community	Due to constraints with family responsibilities, lack of awareness etc. entrepreneurs feel isolated	A community focused online forum where entrepreneurs will have access to like-minded entrepreneurs at any time
Societal Issues	Female entrepreneurs can sometimes be viewed negatively	KPesa should invest in community projects to empower female entrepreneurship beyond their community

FIGURE 2.2.1: Author's analysis of solutions required in the web app to address the challenges faced

KPesa's core business model addresses the barriers to financial capital. However, shifting the mindset from need entrepreneurship to opportunity entrepreneurship requires additional features which may not be required in developed economies. Building an education programme focused on a community-first approach will help organic idea generation while concurrently educating females about the benefits of opportunity entrepreneurship. Approximately 70% of Tanzanian female entrepreneurs have less than four employees (Isaga, 2018); therefore, a focus should be placed on the core competencies of management, hiring, and growing a business. The education programme can be reinforced through a community forum connecting like-minded women, overcoming the barriers associated with networking as they can meet each other and generate ideas in an informal setting. Finally, emphasis should be placed on investing in changing the community's attitude towards female entrepreneurship. This will likely involve reaching communities beyond the platform; therefore, this does not influence the platform's design.

Crowdfunding Business Model

Crowdfunding is a method of raising money to finance projects and businesses, enabling fundraisers to collect money from many investors through online platforms (EU, 2023). There are two categories of crowdfunding: financial return-based crowdfunding and non-financial return-based crowdfunding. Several subcategories exist within the two primary definitions. Non-financial return models include donation and reward-based crowdfunding, while economic return models include equity and loan-based crowdfunding.

Platform Type	Characteristics	Benefits	Risks
Donation	No legally binding financial obligation to benefactors	Less regulatory requirements: low KYC, less sophisticated payment gateways as no return is expected	Transparency of funding opportunities, communication of risks to funders, fraud
Equity Crowdfunding	Investment for stake of ownership	No repayment of capital required, growth capital	Highest risk, investor education, limited due diligence, fraud, communication of risks, vague regulations
Loan Crowdfunding	Capital in exchange for repayment + interest	Lower risk than equity, most common method	Communication of risks, client money management, in-house collection methods

FIGURE 2.2.2: A comparison of different crowdfunding business models

Crowdfunding models often provide financing for a company, charity, or project that cannot access capital through traditional sources. This offers growth opportunities, particularly in nations with underdeveloped capital or credit markets like Tanzania. The penetration of different models varies significantly by region too. For example, Garvey et al. (2017) estimate that loan-based models account for 90% of funding globally, with donation-based models only accounting for a negligible sum. However, in Eastern Africa, it is estimated that 80% of crowdfunding is donation based. There is no available research into why this may be the case; however, examining the risks associated with return-based models, the risk of company failure, lack of due diligence, limited disclosure, and platform risks may result in softer demand for a return-based model in developing nations. Another potential reason, particularly for female entrepreneurs in Tanzania, is that many operate in low-growth industries and face constraints from traditional familial responsibilities (Rutashobya, 2001; Isaga, 2018). This may constrain investor demand due alternative investment opportunities with a higher risk/reward.

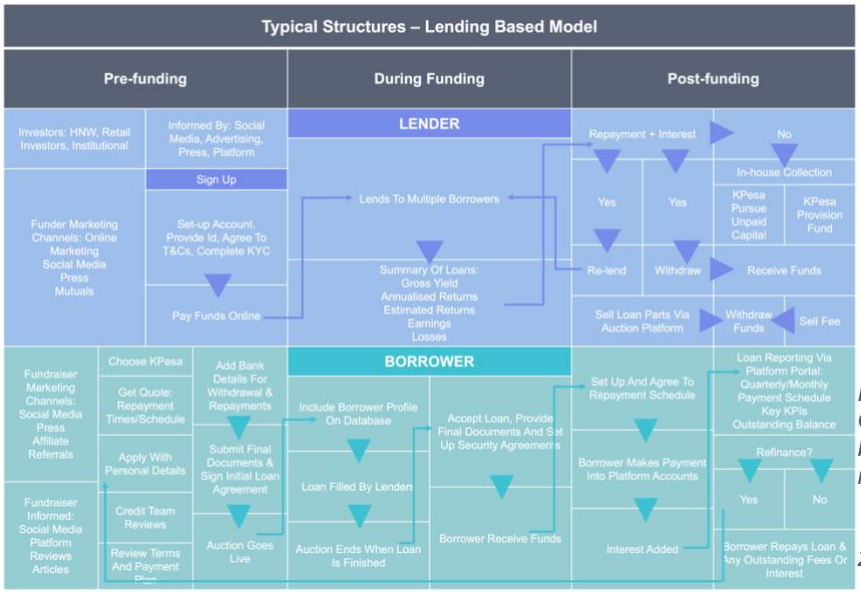


FIGURE 2.2.3: Overview of a Loan-based crowdfunding model.

Source: (Garvey et al., 2017)

An overview of equity and donation-based crowdfunding models has been provided in Appendix F.3. While every model has benefits and drawbacks, a loan-based model would be most appropriate for KPesa. While platforms such as IFundWomen offer rewards platforms, requiring entrepreneurs to post their products internationally may counter the goals of developing a strong domestic economy. KIVA offers a donation-based platform in Africa. However, raising sufficient growth capital for businesses using donations might be difficult. Conversely, loan-based crowdfunding platform is likely the most infrastructurally complex to set up. It requires provisions for loan losses such as in-house collection methods, pursual of unpaid capital and a loan repayment method. The platform must pay out to both the investor and the entrepreneur. KPesa must consider all these factors before deciding its model of choice. For now, when developing the web app, it is most appropriate to build a prototype that can be adapted to any mode they choose to pursue.

Regulatory Context

Currently, there is no regulation specific to crowdfunding in Tanzania and KPesa plan to register as a Microfinancing organisation. This may soon change as a draft regulation has been proposed by the Capital Markets and Securities Agency (CMSA) outlining guidelines for developing and regulating Crowdfunding in Tanzania. The main objective of promulgating a regulatory framework for Crowdfunding operations in Tanzania is to sustain market integrity and investor protection (CSMA, 2022). The key regulatory requirements are indicated in FIGURE 2.2.3.

Regulation	Description
Requirement for license	Crowdfunder must be limited by shares and incorporated in Tanzania
Appointment of Reasonable person	Crowdfunder must appoint at least one person certified by the CMSA as an authorised Crowdfunder, and shall undertake the role of the main contact with the CMSA
General Obligations	(a) Carry out due diligence on prospective issuers (b) Carry out investor education (c) Have AML processes that meet requirements
Due Diligence (DD) requirements	(a) Must conduct extensive DD on issuers. – identity, the fitness of directors, verification of financial statements, business proposal, proof of address, details of incorporation (b) Must conduct DD on investors – proof of address, source of funds, risk assessment
Safeguarding of customer funds	Crowdfunders must safeguard funds by setting up a trust account in a licensed institution and can only be released under certain conditions

FIGURE 2.3.4: Key points from Tanzania's new crowdfunding regulation proposals. Source: (CMSA,2022)

While the regulation has yet to be introduced, KPesa needs to be proactive and meet the requirements in the event of an introduction of regulation. KPesa must register as a company in Tanzania, build a comprehensive policy on AML procedures, and introduce safeguards into the web app to comply with pending regulations. KPesa must also build an education element for the web app for investors and entrepreneurs.

This should educate investors on the benefits of crowdfunding while also highlighting the risks associated with such investments. Finally, once the platform is complete, KPesa must complete an IT & financial audit and provide a written declaration from the auditor providing assurances of adequate IT readiness, and sufficient human, financial and other resources to carry out operations.

3. WEB APP OVERVIEW

This chapter discusses the objectives of the web app, the technical environment and provides a description of the web app's functionality as well as overview diagrams. Additional context from a video demonstrating the functionality can be found in Appendix D.2.

3.1 Web App Objectives

This web app aims to provide the client with a mobile-first web app that allows entrepreneurs to log in, participate in discussions with other entrepreneurs in a forum, and post their campaigns for investors to view.

The requirements of the web app were agreed upon as follows:

- Identify appropriate technologies to develop the web app, making it accessible to users based in Tanzania from their mobile devices. Due to limited resources, there should be a minimal financial outlay for the client. The selection of appropriate technologies is detailed in Section 4.2 – Choosing Suitable Technology.
- Develop a user-friendly experience which is simple in design and easy to use for users with limited technology experience. Further information is detailed in Section 4.3 – Designing the Web App.
- Create a lightweight solution which is designed for environments with low bandwidth.
- Develop a public forum where entrepreneurs can log in and interact with other like-minded entrepreneurs.
- Develop a feature where entrepreneurs can build a campaign seeking investment. The campaigns must then be viewable to the public, who can engage with the profile and share campaigns.

3.2 Technical Environment

The client comes from a non-technical background and did not specify the required languages or frameworks. The only specification was to build a web app. Therefore, I was responsible for choosing appropriate languages and frameworks to build the web app. The web app was developed using HTML, JavaScript, React.js, Next.js, Google Firebase, and Tailwind CSS.

Web App Structure

The web app structure features a front end and a back end. The user interacts with the front end; the languages used on the front end are HTML, JavaScript, React.js and Tailwind CSS. HTML and React.js were used to build components on the front end, and then Tailwind CSS, a CSS framework, was used to style the app. Next.js was used for server-side rendering, which connects to the Firebase API. Firebase is a NO-SQL database used to store the client's data. Further detail on Firebase can be found in Section 4.2 – Choosing Suitable Technology.

React.js

React.js is an open-source front end JavaScript library for building user interfaces based on components. Unlike other languages, when a webpage is requested by typing a URL, React.js uses a unique approach called client-side routing which renders webpages more efficiently. In other languages, when a link is clicked or a new request is made, the entire page is reloaded, which can be slow or inefficient for data-heavy applications (Herbert, 2022). However, client-side routing only requires the specific portion of the webpage that needs updating to be reloaded. React finds the least expensive way to path the document object-model (DOM) with an update without rendering the actual DOM (Herbert, 2022). This results in better performance and a more dynamic user experience. Many popular apps are built using React, including Facebook, Instagram, Netflix, Reddit, Airbnb, WhatsApp Web and many more.

Next.js

Next.js is a framework for server-side rendering. Most frameworks use client-side rendering, which means the data is rendered directly in the user's browser. Client-side rendering can be difficult to keep fast when using a mobile (Miller and Osmani, 2022). Next.js uses server-side rendering or pre-rendering to optimise web pages with constantly changing data. Pre-rendering fetches external data, and React components are transformed into HTML before the result is sent to the client. In a React app, the browser receives an empty HTML shell from the server along with JavaScript instructions to construct the UI. This is client-side rendering. Next.js pre-renders every page by default. Pre-rendering means the HTML is generated in advance, on a server, instead of doing it all on JavaScript on the user's device (Vercel, 2023). The user will see a blank page while rendering in a client-side rendered app. While in a pre-rendered app, the user will see the constructed HTML. Next.js optimises React apps to make them fast, dynamic, and lightweight, which is particularly important in the context of the client.

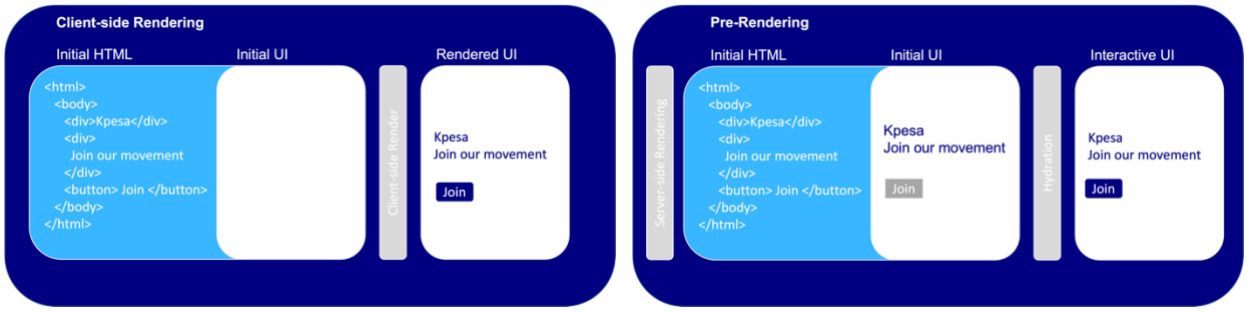


FIGURE 3.2.1: A visual comparison of client-side rendering vs pre-rendering; Source: (Vercel, 2023)

Tailwind CSS

Tailwind CSS is a CSS framework designed to enable the fast creation of apps. Use cases include controlling the colour, layout, typography and much more without leaving the HTML. For example, I created a rounded green button on the login page that dynamically changed to lighter green on hover. It required the following code:

```
<button
  type="submit"
  className={
    "mt-4 mb-3 w-full bg-green-500 hover:bg-green-400 text-white py-2 rounded-md transition duration-100"
  }
>
```

FIGURE 3.2.2: A snippet of code demonstrating how Tailwind CSS seamlessly integrates into code

The button has less custom CSS, reducing the space required and resulting in a more efficient web app.

3.3 Web App Description

The two main components of the web app are discussed in this section. These components are the chat app and the campaigns section. Section 4.4 – Developing the Web App provides a comprehensive overview of the development of each component.

Chat Feature

The chat feature was built to fulfil the client's requirement that the app should have a forum where like-minded entrepreneurs can exchange ideas. The chat has a set of admin-defined channels accessible to all entrepreneurs. Users can navigate each chat heading using a sidebar on the left-hand side. The chat is in full screen on mobile devices, navigating to other chats through the back button, as seen in Figure 3.3.1. The user-friendly chat feature provides an easy-to-use text box where users write messages. The chat updates live in real-time, meaning the user does not have to refresh to view the most recently received messages; the technical details of chat feature will be discussed further in Section 4.4. – Developing the Web app. Figure 3.3.1 shows examples from the output of the chat feature. Appendix D.2 shows further examples of the output to viewers on a larger screen. Section 4.3 – Designing the Web App contains the chat app's design details.

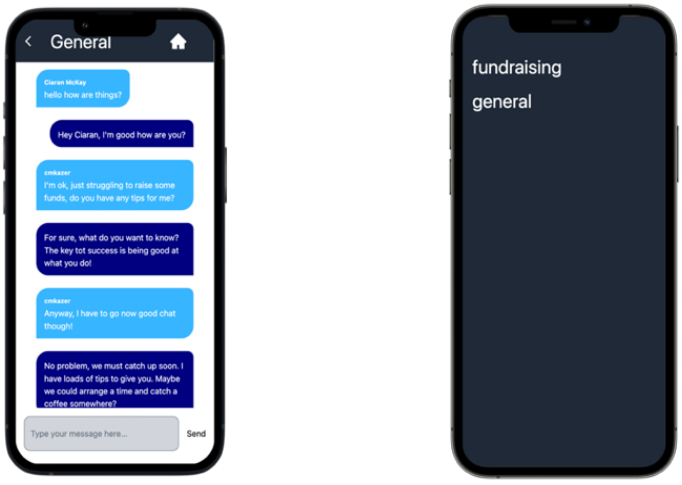


FIGURE 3.3.1: An example of a mobile user's experience in the chat feature.

Campaigns Section

The campaigns section is broken into two parts: creation and viewing. The campaign creation page is available to entrepreneurs who are seeking investment. The user enters the campaign details in a multi-step form, generating a campaign. From the investor perspective, the campaign is then populated in the web app, where they can view a gallery of campaigns and choose individual campaigns that interest them. They can pledge to invest or share the campaign with others. Figure 3.3.2 displays a screenshot of the current user journey. For more information on the different user groups in the app, please read Section 3.4 – Web App Overview Diagram.

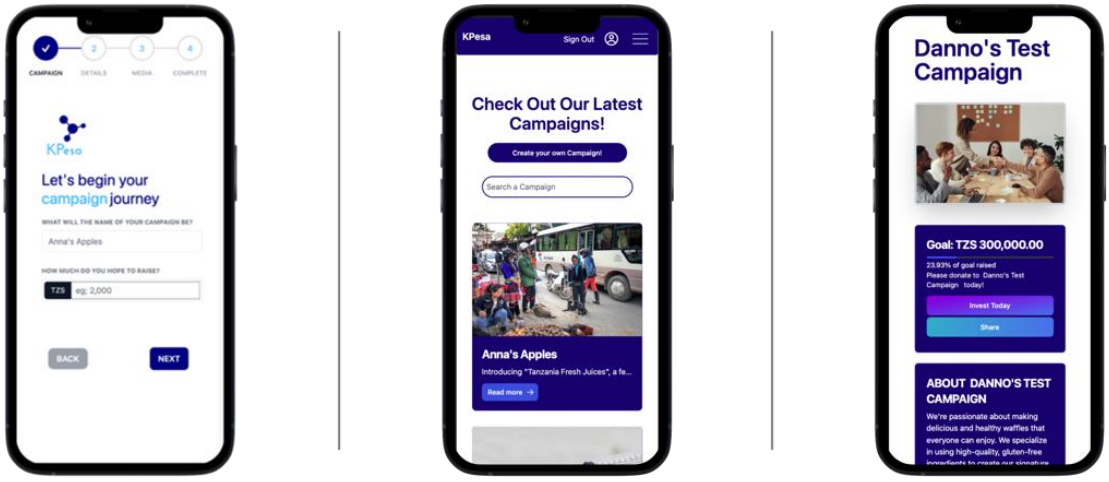


FIGURE 3.3.2: A user's experience of a campaign

3.4 Web App Overview Diagram

The web app built is a full-stack web app that uses Firebase's 'Firestore Database'. As mentioned, Firestore is a No-SQL document database queried using the server-side JavaScript Framework Next.js and custom Firebase methods. The Firestore database stores all the information on our users, campaigns, chats, and authentication data. The outline of how each entity interacts with the app is outlined below:

- User: The user is anyone who uses the web app. The user can sign up or log in to the database. There are three types of users: entrepreneurs, investors, and non-registered users.
    - Entrepreneurs can access the forum, post campaigns, view campaigns and pledge to invest in campaigns.
    - Investors can view campaigns and pledge to invest in campaigns.
    - Non-registered users can view campaigns but cannot pledge to invest without signing in.
- Database: The Firestore database contains all the necessary information data to run the web app. It is comprised of four collections: Campaigns, Channels, Messages, and userData. An authentication collection is also used to store a user's login details.

Please read Section 4.3 - Designing the Web App to read more about the design of the database.

The development required full-stack engineering including building a front end and a back end. The user interacts with the web app at the front end, and the back end acts as the app's engine. The back end handles user input and queries when using the app; this is not visible

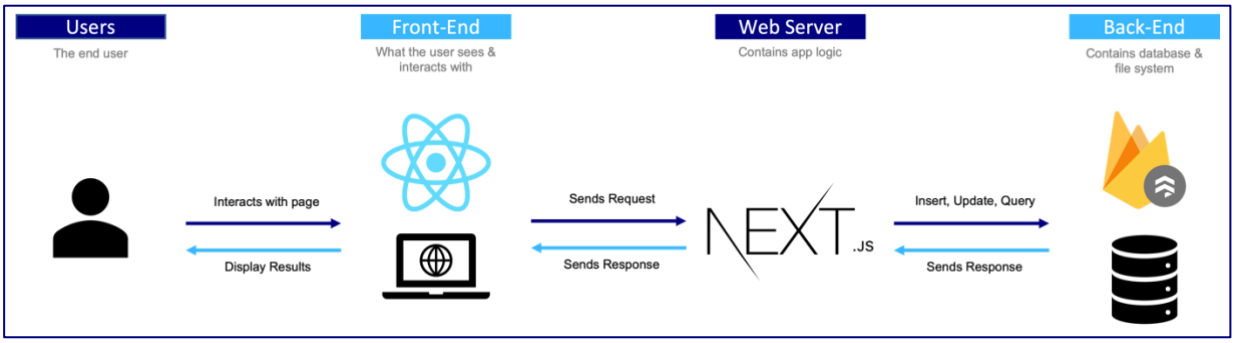


FIGURE 3.4.1: Shows the interaction between different entities in the web app

to the user. For example, when a user builds a campaign, they insert details such as the campaign name and target sum to raise. When submitted, this is entered into the database. When a user requests to view the campaigns, the back end takes the user request, queries the database, and retrieves the relevant information. Once retrieved, this information is passed to the user and displayed on the front end. Figure 3.4.1 shows the interaction between a user and the web app from front end to the back end.

Figure 3.4.2 portrays a sitemap for the web app. There are three different levels of authorisation in the web app: public, investor and entrepreneur. Entrepreneurs have access to every level of authorisation, while investors can only access public pages and pages with investor access. A description of the different components in the web app can be found in Section 4.4 – Developing the Web app.

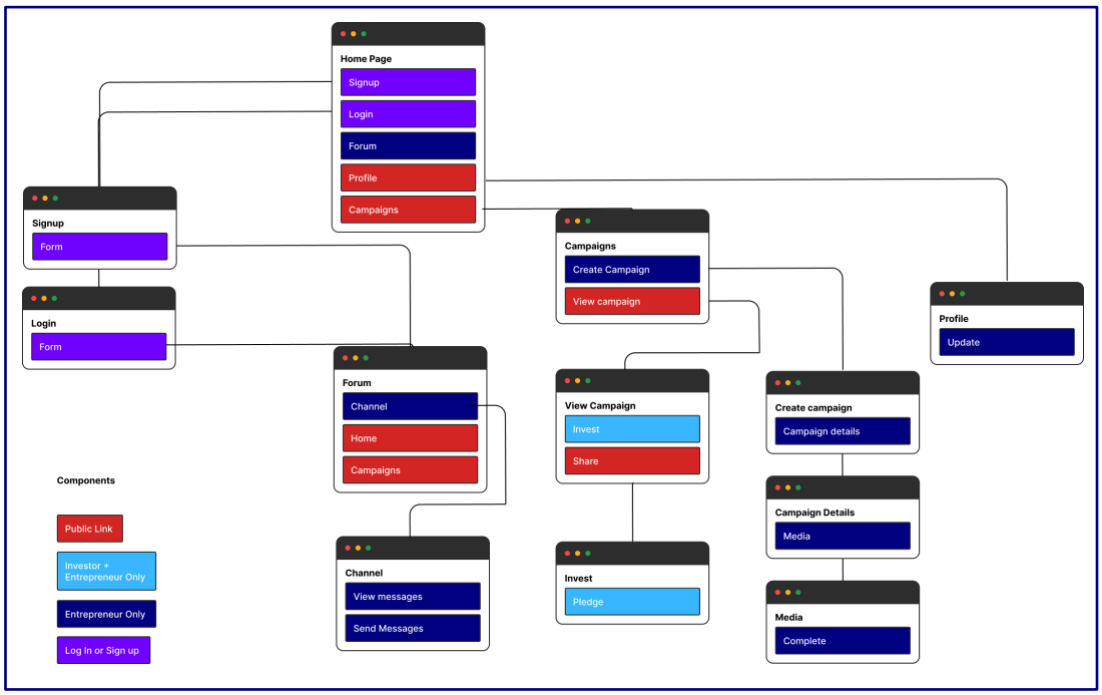


FIGURE 3.4.2: Sitemap

4. BUILDING THE WEB APP

The following chapter outlines the work completed to build the web app. An agile development methodology using a Kanban framework was implemented to develop the web app successfully. The client's inexperience required a flexible approach to development. This allowed me to suggest features and lead the project, testing my developer and product management skills. Further detail on my approach and the methodologies used is found in Appendix C. The project was broken down into the following phases:

- Defining the web app requirement
- Planning the approach
- Designing the web app
- Developing the web app
- Testing the web app

} Iteratively

4.1 Defining the Web App Requirements

The early stage of the project required understanding the features required to meet the needs of the entrepreneur. A highly collaborative approach was taken, combining the expertise of the founder with the research gathered in Section 2 to define the web app requirements.

As discussed in Section 2.2, a community-first approach would be KPesa's USP, inspiring idea fruition. To meet this requirement, it was agreed to develop a chat feature enabling users to participate in public forums and interact with each other, fostering a community. It was essential to ensure the web app is lightweight and optimised for low-bandwidth areas and the user interface is intuitive, prioritising a mobile-first approach. Further discussion regarding the development of the chat feature can be found in Section 4.4 – Developing the Web app.

Following the development of the chat feature, it was decided to integrate a campaign section into the web app. The crowdfunding model of KPesa is still undecided, so a generic campaigns approach was built as a foundation for KPesa to adapt at a later point. Drawing inspiration from 'IFundWomen', 'GoFundMe' and 'iDonate' it was agreed to build a campaign section where entrepreneurs could launch a campaign including details such as their target, a campaign description, and related media. It was also required for to enable investors to view active campaigns.

4.2 Choosing Suitable Technology

As the client had limited technical expertise, it was essential to conduct thorough research into choosing the proper languages, software and tools which would be used to develop the web app. An essential requirement was for the website to be lightweight as it would be deployed in regions with low-bandwidth internet connectivity. Several factors were prioritised in this decision, including the efficiency of every programme, ease of use for non-technical users and how intuitive it would be to implement.

Proprietary Chat Feature

Various solutions were considered to determine the best web app development approach, including off-the-shelf products like Slack or Discord. After careful consideration, I built a proprietary chat feature for two main reasons.

First, given the platform's user base, it was essential to keep the web app lightweight. Popular chat apps have many features, but they are also data intensive. For example, the Discord and Slack apps require 169 MB and 329 MB of data to download (IOS App Store, 2023). This could pose problems in an area with limited data, ultimately limiting the platform's adoption.

Secondly, by integrating a proprietary community feature into the web app, entrepreneurs can connect with other female entrepreneurs, learn from each other, and develop trust. Ultimately, this would help develop a community, a key requirement found in the exploration stage of the project. Therefore, it was deemed most appropriate to build a proprietary feature.

Firebase

Firebase's Firestore is a NoSQL document database deployed in the project to facilitate automatic scaling and deployment. In contrast to traditional SQL-based relational databases, it does not use tables and rows but instead stores data in documents organised into collections. Documents contain key-value pairs and are stored in collections as document containers.

One of the advantages of Firestore is its scalability, which enables it to efficiently handle large volumes of data and support high levels of simultaneous user traffic (DatabaseTown, 2023). Firestore is a schemaless database that confers complete freedom regarding the fields included in the documents. Document databases like Firestore have simple APIs that make integration more accessible and may require less specialised expertise than relational databases. Consequently, friction should be reduced when handing over to a new developer. A drawback however is that document databases may exhibit limited querying abilities when performing complex multi-table joins (DatabaseTown, 2023).

Source Code Editor

Visual Studio Code 2, a code editor, was utilized to develop the web app. This free code editor offers native support for Node.js development with JavaScript. It also boasts many powerful tools geared toward web technologies, such as React, HTML, CSS, and JSON.

A significant benefit of Visual Studio Code 2 is its highly intuitive Intellisense feature, which can detect incomplete code snippets and automatically declare comment syntaxes. Additionally, this code editor seamlessly integrates with Git, facilitating efficient project version management and enabling users to retrieve or save instances to the repository easily.

Web App Deployment Using Vercel

The web app is deployed on Vercel, a cloud platform which maintains the Next.js web development framework. Vercel was chosen due to the generous usage limits on the free account and an in-depth website performance analysis, allowing for efficient testing of live features on desktop and mobile. The current deployment can be found at kpesa.vercel.app.

4.3 Designing the Web App

The design phase involved two primary stages: the front end user interface design and the back end web app design. The design process was guided by usability and efficiency principles to meet the client's needs.

The project's primary objective was to create a web app providing entrepreneurs a convenient platform to join a community, post campaigns and engage with others. Hence, the web app's design focused on ensuring it was intuitive and user-friendly. The design process was divided into two key sections to achieve the project's goals: the database and user experience design. Both sections were meticulously approached, with attention given to every detail.

Back End Design

The back end design for the web app involved several phases, including authentication, chat design, and campaign design. As KPesa is community focused, it is essential that users can log in and interact. Firebase offers native support for authentication requests, providing secure user authentication. Using Firebase, a sign-in with Google and a custom email and password sign-in option was integrated.

In contrast, SQL authentication often requires a more complex setup, which can be challenging for developers without advanced knowledge of databases. Additionally, SQL-based authentication does not offer third-party authentication options. Therefore, the easy integration of Firebase authentication further enhances its suitability for the project/ Authentication was developed by using a *useAuth()* hook. Hooks allow developers to use state and other react features in functional components. Custom user-defined functions were created in *useAuth()*, such as *"signup",* seen in Figure 4.3.1, which allowed Firebase to authenticate the user easily.

```
const handleSubmit = async (event) => {
  event.preventDefault();
  if (password.value == password_confirm.value) {
    try {
      await signUp(
        email.value,
        password.value,
        userType.value,
        firstName,
        lastName
      );
      router.push("/channel/ZHLL1uu44KdB3v9iaxXN");
    } catch (error) {
```

```
const signUp = async (email, password, userType, firstName, lastName) => {
  try {
    const response = await createUserWithEmailAndPassword(
      authorisation,
      email,
      password
    );
    const user = response.user;
    await sendEmailVerification(user);
    setUser(user);
    await setDoc(doc(db, "userData", user.uid), {
      userType: userType,
      firstName: firstName,
      lastName: lastName,
    });
  } catch (error) {
```

FIGURE 4.3.1: Sample code used for authentication

A valuable feature of document databases are hierarchical relationships. Two collections were used to store the chat's data. The first collection channels contain the Channel Id and name. A sub-collection was created for each channel, which contained the messages associated with that channel. Hierarchical relationships enable efficient querying as you do not need to search the entire database. An example of this data can be seen in Figure 4.3.2.

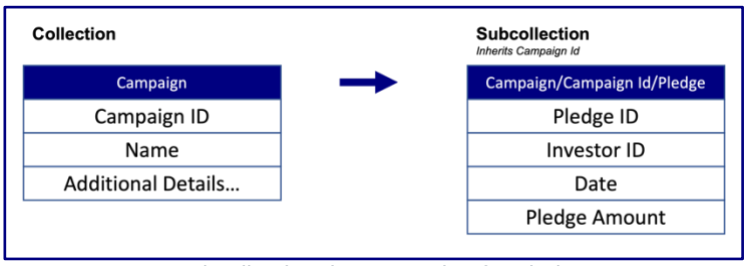


FIGURE 4.3.2: Subcollection demonstration for pledges

The planning phase of the campaign builder entailed various adaptations of the existing infrastructure, to create an immerse interface for campaigns. A campaign collection was created. To personalise the campaign, it was also necessary to include a photo. However, it was not possible to directly upload a photo to the collection. The solution was to store the photo in a 'Firebase Storage' folder. Once the upload was complete, a function was triggered to retrieve the download URL from the Firebase API and add it to the campaign collection. When a user wanted to view a campaign, the backend would make an API call to Firebase using the stored URL to fetch the data. Finally, campaign pledges were also stored using hierarchical relationships as a sub-collection of the specific campaign.

Entity relationship diagrams, data flow diagram and the relational scheme can be found in Appendix C.

User Interface Design

A user-friendly interface ensures a positive user experience when using a web app which can be achieved in two ways, first through interface design and second through the efficient flow of the website.

Although KPesa has a designer as part of their founding team, they were unavailable during the project timeline. Initially, designing the user interface was challenging, and some mock-up creation was introduced using Figma, a design tool. The mock-up served as a visual reference while building the app. An example of a mock-up can be found in Appendix C.3. Combining KPesa's branding, inspiration from similar sites, and the mock-ups ensured a clean, consistent, and intuitive design across the web app. Sample screens from the web app can be found in Appendix D.2.

Activity Flow

When the database and design planning had been completed, designing a user activity flow was essential to bridge the gap between the front and back end. From the site map seen in Section 4.3 – Designing the Web app, there are many possible routes a user can take, and different user groups will experience different activity flows.

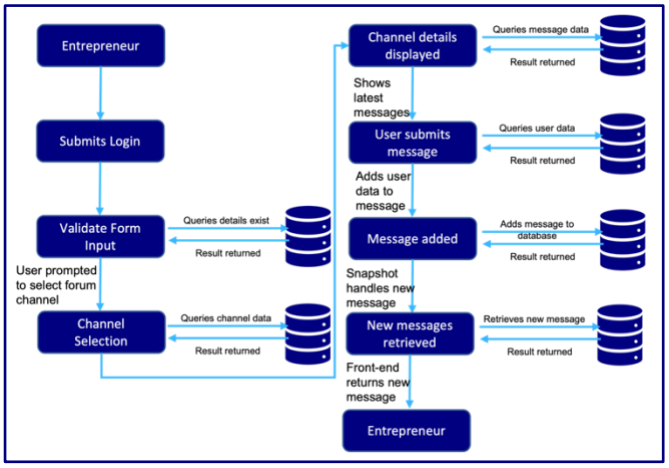


FIGURE 4.3.3: Activity Flow Diagram for an entrepreneur logging in

Using the entrepreneur as an example, we can analyse the activity flow on login. When designing the activity flow, it was essential to approach it from KPesa's community-first mindset. Therefore, the web app was designed to redirect the entrepreneur to the forum upon login. The diagram in Figure 4.3.3 indicates the journey undertaken by a user when they wish to use the chat feature. Additional activity flow diagrams can be found in Appendix C.

4.4 Developing the Web app

This section details the stages and iterations involved in the development of the web app. The web app was developed under the Agile methodology, mixed with Kanban boards. Development was completed iteratively alongside the designing of the web app. There were two primary stages: the development of the chat feature and the campaigns section. The source code of the web app is available on GitHub in Appendix D.1.

Chat Feature

The chat feature was the first stage of development. The chat connects users in a safe community-based setting. The first approach was to develop the database and design the website using React.js and Tailwind CSS. Finally, the front end and the back end were merged using Next.js. The user experience is fragmented depending on the device being used.

- A mobile user views a condensed screen. First, they are directed to a screen which shows the various channel options. Once they click a channel, they view it in full screen, with the navbar only displaying a back button.

- A desktop user is directed straight to the forum, where there is a sidebar to navigate through channels, reducing the required steps.

The chat feature was created using three components *Channel.js, Channels.js* and *Message.js* and one page called *[id].js* inside a folder called Channel.

Message.js takes a message object as a prop which contains properties such as the user id, text, and the time the message was sent. The *useAuth* hook checks which user sent the message. The component returns an individual message bubble.

Channel.js renders the details of the channels which are available to chat in. This was done by returning a list of items to be compiled with other channels. Each list item is a dynamic URL path using Next.js that uses the unique id of the channel as a parameter value, meaning that the URL will be different for each channel. This parameter value is inserted dynamically when the link is rendered, which lets the user easily navigate to the desired channel.

The Channels.js component is set up to return the list of channels wrapped in a styled class. The *useEffect* hook is used to run a query to fetch the channel names. The *getDocs* method fetches the initial set of documents from the database. To ensure the client-side remains up to date, a listener method called *onSnapshot* is set up to listen for any changes to the channels collection and update the user interface. A snapshot handler is then used to create an array of the channels using a key pair value of “id” as the key and the data as the values. The result was a div created where the list of channels was available.

The *selectChannel.js* page allows channel navigation for mobile users. *selectChannel.js* calls on the channels component and displays the channel drawer in full screen on a mobile.

[id].js is a page that renders a specific channel. This page retrieves and displays messages posted in a specific channel and provides navigation tools. The channel.js, Channels.js and messages.js were used to compile the page. When a user sends a message, a handle submit function is triggered adding a new message to the messages document in the database.

First, the function prevents a refresh by calling on *the preventDefault* function, and then it uses the *addDoc* method, which adds a new document to a collection. Each time a message was added, a *scrollIntoView* method would return the user to the latest message sent in the channel.

Next, it was required to query the messages. A query function is used to find the messages in the collection for the channel Id. The messages are then sorted in ascending order by date created and placed in a data array. A *getDocs* method and an *onSnapshot* listener are used to gather the initial messages and update the UI. The *onSnapshot* listener is critical in a messaging app as it fetches data from the server and updates the client’s UI without refreshing the page, allowing for seamless retrieval of new messages.

Once the database interactions were set up, I could design the chat. The chat is optimised for both mobile and large screens. For familiarity, a style like WhatsApp Web is used. Therefore, using CSS, a flex-column layout was set up on a large screen to allow the screen to be split between ¼ channels and ¾ chat. The channel drawer is hidden on smaller screens, and a small back bar is used for navigation. The page calls on the various components, such as *Channels.js,* that were built. The benefit of using these components is that they act like Lego bricks which can be reused in any part of the app, thus reducing the amount of duplicate code, making a more efficient, faster app.

Finally, a protected route component wraps this page so that if a user that is not an entrepreneur tries to gain access to this page, they will be redirected to the login page.

Campaigns

The campaigns section was the second primary stage of development. The objective for this page was to create a flexible solution that can be easily modified once KPesa have finalised their crowdfunding model.

There are four sections to the campaign section. First, entrepreneurs needed to be able to build a new campaign; this was done using a multi-step form using an array of components called *campaignDetails.js, Details.js, Media.js, and Final.js.* The form was controlled by a stepper called *Stepper.js* which handles the user's form progression. Once the campaign was built, a campaign viewing section was required. This was done by first creating a component called *Campaign.js* that retrieves data from the database and populates the data. Then it is passed into a component called *CampaignList.js* which creates a gallery view of the campaigns for investors to view.

Finally, once the components were all built, a page called campaign/[id].js was built to return the data to the front end. The [id].js file used multiple components to create a multi-step page that allowed logged-in users to pledge donations to the founders. Further detail on the build is provided below.



```
//step handler
const displayStep = (step) => {
  switch (step) {
    case 1:
      return (
        <UserCampaign
          userData={userData}
          setUserData={setUserData}
          handleClick={handleClick}
        />
      );
    case 2:
      return (
        <InvestorDetails
          userData={userData}
          setUserData={setUserData}
          handleClick={handleClick}
        />
      );
```

```
const handleClick = async (direction) => {
  let newStep = currentStep;

  direction === "next" ? newStep++ : newStep--;
  //Check if steps within bounds
  newStep > 0 && newStep <= steps.length && setCurrentStep(newStep);
  console.log("userData", userData);

  if (currentStep === 4 && direction === "next") {
    try {
      console.log("userData", userData);
      const reviewRef = doc(
        collection(db, "campaign", userData.campaignID, "pledges")
      );
      await setDoc(reviewRef, {
        id: reviewRef.id,
        campaignID: userData.campaignID,
        investorId: userData.userID,
        pledgeAmount: userData.pledgeAmount,
        investorAddress: userData.address,
        investorMessage: userData.message,
      });
      setCurrentStep(newStep);
    } catch (error) {
      console.log("Error", error);
    }
  }
};
```

FIGURE: 4.4.1: Sample code showing the multi-step page (left) and how the stepper is handled (right) in [id].js

The campaign form builder form is a multi-step form broken into four sections; the campaign details, such as the name of the campaign and the investment goal; the campaign information, which contains a description of the campaign, the campaign media, which contains a photo of

the campaign and then finally a confirmation page which provides proof of submission. The form is controlled using a stepper called Stepper.js.

The *CampaignDetails.js* component handles the initial details required to build a campaign. Next/image was used to import KPesa's logo as it is a pre-optimised image component that provides better image loading performance in a Next.js app. It provides automatic image optimisation and lazy-loading out-of-the-box, reducing the page load time. The component captures the input using a handle change function and then passes the data held in *userData* through to the next stage as props.

The *Details.js* component is used to capture the description of the campaign. The formatting was identical to *CampaignDetails.js,* except a *textarea* HTML element is used instead of an input element because an input element only accepts single-line input. In contrast, a textarea element can accept multi-line text input. In this instance, it was deemed more appropriate.

The *Media.js* component allows the user to upload a campaign image. The layout is the same; however, the data handling was different as uploading media to a collection is impossible. The Firebase Storage feature is required. An event listener is used to track the upload; once the upload is finished, a call-back function is used to retrieve the download URL from the firebase API. The *userData* prop is then updated to include the URL. The confirmation button triggers a function which retrieves the data from *userData* and adds a new document to the campaign's collection. The user is redirected and greeted by a congratulatory message.

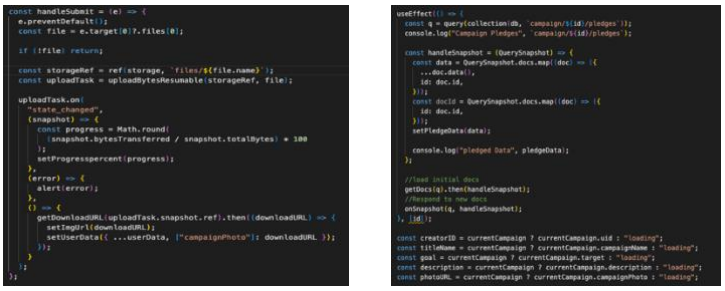


FIGURE 4.4.2: An example of how image upload (left) and retrieval (right) is handled in campaigns

The campaigns were populated using two components: *Campaign.js* and *CampaignList.js.* The setup was like the channel list. *Campaign.js* was used to populate a single card displaying the campaign details. A single prop called campaign was passed into the component, then deconstructed to extract the campaign details. The card was designed to display a campaign preview with a dynamic link to the specific campaign URL. An ellipse was used, hiding the overflow text from the preview to ensure a clean, consistent look across the campaigns.

CampaignList.js populates a list of the campaigns in a gallery view by important the *Campaign.js* component and querying the database to build an array of the campaigns and their details using a *useState* hook. The campaigns array is mapped using the *Array.prototype.map* method, returning a *Campaign* component for each campaign in the array. The key prop is set to campaign id so React can track the components as they update.

campaign/view.js provides a gallery view of the live campaigns on the client side. The page provides several options, such as navigation or viewing the campaigns. Navigation options were included using the Navbar and Footer components, and the campaigns were rendered using the *CampaignList* component. The database authorisation prop is passed into *CampaignList* enabling data retrieval from the database. The gallery list of each card is then populated using a "flex" layout property.

campaign/[id].js is used to view individual campaigns. If a user clicks on a campaign in the campaign viewer, Next.js will take the key to the campaign and pass it as a dynamic URL which links to campaign/[id]. The data was fetched for the campaign from the database using a query. To prevent errors, ternary operators displayed "loading" if the data had not been fully retrieved when the pages loads, as seen in Figure 4.4.2. This is an essential error-handler in low-bandwidth environments where fetching data may take longer. The page has an intuitive design. It is split into two columns, the campaign photo and description to the left and 1/5th of the page on the right-hand side dedicated to a card displaying the campaign's goal, an option to donate or share and then information about the creator. In mobile view, this changes to provide a single column of information, offering a better experience.

The user has the option to invest in the campaign or share the campaign. If the user selects to invest, a pledge form is displayed using a *displayStep* function and three props *userData*, *setUserData*, and *handleClick*. The pledge form consists of three stages, personal details, pledge details and a pledge review. On confirming the pledge, the *handleClick* trigger retrieves the *userData* prop and utilises the *setDoc* method to add the pledges to a campaign's subcollection. The campaign view uses the *handleSnapshot* method to retrieve the pledge data and update the progress tracker. The progress tracker is created using the *reduce* method, which iterates over the *pledgeAmount* array and accumulates the value. The value is then converted into a percentage of the total goal and a progress bar is displayed.

If the user chooses to share the campaign, a modal pop-up appears in the centre of the screen, which displays options to share on Facebook, WhatsApp, Messenger, and email or copy to clipboard. The "React-Share" library created the shareable links with a custom message. For example, clicking the email share button opens the default email provider and creates an email with the subject line, "Check out this awesome campaign on KPesa, a crowdfunding platform for female entrepreneurs", and includes a link to the campaign in the email body.

General Web App

To support the app, other pages were required to be built during the development of the app. This includes a login page, a signup page, a navigation bar, a footer, and a profile section where users can update their profiles. Further details on authentication can be found in Section 4.3 – Designing the Web app.

The footer provides quick links to other sections of the web app to assist the user and is included on most pages. The navbar appears on every page. A hamburger option was chosen for its simplicity, providing a clean look to the app, especially on a small screen. A login/logout button is also included, which dynamically changes depending on the user's authentication status. This is checked by calling on the *useAuth* hook and checking whether the user is authenticated. A *ProtectedRoute* component was also created to prevent users from accessing pages they are not authorised to access.

4.5 Testing the Web app

The web app was created using the agile development method. This means that testing occurs after the development of each feature in the web app. This ensures a proper and working solution for the client before deployment, and errors are reduced when launched.

Many testing types can be used, such as unit, integration, functional, performance, and acceptance tests (Pittet, 2023).

For this project, unit testing was used, followed by integration testing. Unit testing refers to low-level testing, which involves testing individual methods and functions of the software's classes, components or modules (Pittet, 2023). Integration testing typically takes place at a later stage than unit testing. The purpose of the integration test is to verify that different modules of the app work well together. These tests are more sophisticated as they require multiple parts of the web app to interact. Finally, the app was deployed on Vercel to examine the app's behaviour on different devices and browsers.

KPesa will soon begin testing on a group of beta testers in the early Summer. Three entrepreneurs have committed to the trial and several investors have expressed interest. The goal of the test will be to collect data on the user experience. I will be providing support to ensure a smooth transition. Additional details can be found in Appendix G.1 – Next Steps. Once testing is complete, the web app will need to be translated into Swahili to be compatible with KPesa's target market.

During the testing phase issues arose when utilising the preferred method for rendering images using Next/Image. Initially, the HTML <img> tag was required due to Next/Image's default disallowance of web links. The Next.js image optimisation behaviour was modified via the next.config.js file to specify remote sources for image optimisation to overcome this problem. In this case, images stored in Google Cloud storage were identified by the protocol, hostname, and pathname specified in the remote Patterns array. This configuration ensured that images from the specified Google Cloud Storage bucket were optimised during the Next.js build process. To read more about why Next/Image is used for image rendering, please read Section – 3.4 Developing the Web app.

Another issue that arose was related to the route directory. As the chat was the first feature created, the default directory was the chat. This meant that investors or non-registered users

would immediately be directed and presented with an error, causing an unpleasant user experience. This was not noted until the web app was deployed, as the testing on localhost failed to simulate the real experience accurately. This required me to change the default directory to the login page. Vercel's real user experience score, a quantifiable measure of how pleasant the web app is to use, jumped from 59 to 99 immediately following the correction. Further details on testing conducted can be found in Appendix E - Testing.

4.6 Challenges Encountered

Client Engagement

The client, KPesa, was very supportive. However, some challenges arose from engagement. The client was in Zanzibar for three months; engagement was therefore challenging as the client had limited access to quality internet. The client is also a first-time founder with many blue-sky ideas. This created an exciting task; however, they were still determining what features they would like from the web app, ultimately leaving me to drive the features and design the app. This drove the initial scoping investigation phase to gain an understanding of the need for features. I also adopted a Kanban-style priority list where they would send me a list of features they would like to achieve, ranked in order of priority. This, along with the return of the founder to Ireland in February, helped set more explicit guidelines for how I approach the web app.

Creating an Efficient Web App

A unique aspect of this project was the requirement to build a web app that offered efficient performance to the user. In Tanzania, an important consideration is the data required and the speed of the web app. To ensure efficiency, the selection of appropriate technologies and the data flow was designed carefully. The Vercel deployment's performance analytics was used to analyse the performance of the app which provided quantitative scores on the time to load each page. One adjustment needed was to reduce the size of the campaign photos and implement Next/Image, which helped with more efficient rendering. An example of Vercel's dashboard can be found in Appendix E.2 and the rationale behind technologies can be found in Section 4.2 - Choosing Suitable Technology.

Technical Experience

Despite completing modules in Java in first and second year, I had not completed a software engineering project and have no experience building a front end. Therefore, choosing the right language and building a full app was a steep, but rewarding learning curve. To prepare myself I first completed extensive research on the best technologies to choose, which can be found in detail in Section 3.2 – Technical Environment and Section 4.2 – Choosing suitable Technologies. I then completed a LinkedIn Learning course in full-stack engineering. When my requirements were settled, I felt confident to undertake the task of building the web app.

5. CONCLUSION AND RECOMMENDATIONS

This section discusses the conclusions the author has reached and recommendations to the client on areas of further development and improvement.

5.1 Conclusion

The project's objective was to investigate how a crowdfunding app should be designed to address the motivations, drivers, and challenges of female entrepreneurship in Africa. This required applying the information gathered to build a proprietary chat feature where entrepreneurs can engage in a community setting and a campaigns section where entrepreneurs could launch campaigns and investors invest. The project successfully met the terms of reference outlined in Section 1.3 – Terms of Reference. The app offers functionality where entrepreneurs can engage with each other via a community forum. The app does not offer real funding due to regulatory limitations that must be addressed first which are discussed in Section 2.2. However, I added a pledging web app to enhance its functionality of the web app. This feature enables potential investors to pledge money to support the entrepreneurs. The web app acts as a ready to use foundation for KPesa once their business model is defined and regulatory requirements are met. KPesa have indicated their satisfaction with the output, as seen in Appendix G.2 – Client Feedback.

The project followed a structured approach, comprising a planning, research, design, development, testing, and a deployment phase. While implementing the web app, several challenges were faced, such as the need to learn new programming languages and the difficulties of communicating with a client with limited internet access during the initial phases of the project. Nevertheless, we overcame these obstacles using effective software development techniques, including an agile development framework, regular communication, and skilful guidance from my supervisor. Overall, by leveraging these methods, I was able to successfully navigate through the project's complexities and deliver a solution that satisfied the client.

5.2 Recommendations

While the project was successful in meeting its requirements, as this is an early-stage prototype, many improvements can be made. This section describes some of the next steps required to launch the product successfully. A further discussion of recommendations for KPesa's web app can be found in Section 2.2.

Building a Payments Web App

Building a crowdfunding platform requires a more sophisticated payment solution than conventional payment processing. This is because know-your-business (KYB) and anti-money laundering (AML) checks are required, as well as e-wallets to escrow campaign funds and scheduled payouts. The structure of KPesa's solution will determine the most appropriate

solution. There are three primary categories for crowdfunding: donations, crowd-equity, and crowd-lending. For donations, less KYC is required; however, if it is a crowd equity or crowd lending platform, the functionality of payment handling must be advanced, and compliance checks are essential. In Tanzania, a licensed institution requires a trust account (CSMA, 2022). A breakdown of leading payment gateway providers is included in Appendix F.2.

I have identified several potential solutions; however, none are complete end-to-end solutions. MangoPay is a payment gateway that provides ready-to-use functionalities such as escrows, e-wallets, and native KYB. The platform is built for KPesa's business model. However, unfortunately, they do not offer a payout to mobile money accounts, which may inhibit KPesa's accessibility.

Local payment providers such as Tingg and Flutterwave, who specialise in payment gateways for Africa, offer mobile money payouts at reasonable rates. However, neither platform is optimised for crowdfunding and does not offer KYB services. A more complex solution from KPesa would be required to safely ringfence customer assets and ensure no fraud risks on the platform.

One such solution I have identified which may be optimal is Stripe Connect. Stripe Connect offers a proprietary crowdfunding feature and KYB at low rates. However, it does not offer a payout to mobile money accounts. I have found a wraparound solution to use Zapier, a workflow automation tool that can integrate with Stripe and Flutterwave (Zapier, 2023). A trigger can be set up so that once the action is completed, a transfer is made to Flutterwave, which can then trigger a payout to customers using mobile money. While this solution has yet to be tested, Zapier offers a variety of task automation integrations between Stripe and Flutterwave. The result is that the investor has the peace of mind of paying through the world's leading payment gateways, and the entrepreneurs can receive their capital in the form of mobile money.

Protecting Customers

There are several risks associated with crowdfunding such as the risk of a loss of investment, fraud, and misappropriation of funds. One of the biggest challenges with a crowdfunding platform is preventing fraud. KPesa plan on manually screening user applications, however as they grow, they must automate this process. This includes document management. Services like Inscribe.ai offer fraudulent document protection for use cases such as KYC, account opening, personal lending, and underwriting (Inscribe, 2023). Using a secure third party could help KPesa grow securely.

A requirement from the CMSA (2022) draft regulation for crowdfunding is to provide investor education. It is recommended KPesa build a pre-registration survey to inform investors of the risks associated with crowdfunding. This is an essential step to adequately inform users about risk. Controls should be put in place so a user cannot invest until they have met the necessary KYC requirements and have completed the investor education programme.

TRINITY COLLEGE DUBLIN
Management Science and Information Web apps Studies
Project Report

KPESA
Female Entrepreneurship in Tanzania: Developing a Crowdfunding Solution

14th April 2023

APPENDICES

Prepared by: Ciaran McKay ***Supervisor: Paula Roberts***

A. ORIGINAL PROJECT OUTLINE

Client KPesa: Crowdfunding Platform

More details about KPesa: https://tinyurl.com/m68du6cb

Client background

KPesa is aiming to use the power of Crowdfunding Platform to bridge the financing gap for Women in Micro and Small Businesses (MSMEs). We will raise finance from the public to provide low interest loans to women lead MSMEs.

Women who are unable to access bank loans will be able to access loans for their businesses on our crowdfunding platform. With Africa being the only continent in the world where female entrepreneurship outnumbered male (World Economic Forum). There's a great potential to empower this demographic to eradicate poverty in the continent (UN SDG 1).

Our solution will also promote economic growth by helping women in the informal sector to grow their businesses to the formal sector - Increase a tax base for the Government. We also aim to tackle gender inequality (UN SDG 5) by providing access to finance to women and empower them with financial independence.

Project background and Requirement

User friendly Web app crowdfunding platform

Onboarding questions for credit scorecard with neuroscience element

What is involved for the students?

Skills including: programming for developing the webapp

UX/UI or user friendly

analysis to analyse the data

neuroscience to come up with credit scorecard onboarding question

B. INTERIM REPORT

Management Science and Information Web apps Studies Project:
Building a crowdfunding platform for female entrepreneurs in Eastern Africa
Client: KPESA
Student: Ciaran McKay
Supervisor: Ms. Paula Roberts

---

1. Review of Background and Work to Date

1.1. The Client Company

KPESA, the client is aiming to use the power of Crowdfunding to bridge the financing gap for women in micro and small businesses (MSMEs). Africa is the only continent in the world where female entrepreneurs outnumber men, yet many remain unbanked and cannot access loans through traditional financial institutions (Dushime, 2022). KPESA's solution will promote economic growth by raising finance from the public to provide low interest loans to women lead MSMEs, helping women grow their business from the informal sector to the formal sector.

1.2. Review of Work to Date

The project is a web development project focused on creating an MVP for the client. To do so I will use software languages such as HTML, CSS, JavaScript, Node.JS, React and SQL. I have spent the first phase of the project familiarising myself with full stack development by undertaking a 'Become a Full Stack Developer' LinkedIn Learning Course. I have also conducted extensive research on the funding gap in Eastern Africa and the competitive landscape of crowdfunding platforms to appropriately assist the client.

I have met the client on two occasions, the first an introductory session and the second a meeting to discuss the features desired by the client.

2. Terms of Reference

The finalised terms of reference have yet to be agreed by the client however, a target date of December 9th has been set. The preliminary terms of reference are set out below:

1. Research the funding gap for female entrepreneurs in Eastern Africa and the current solutions available to overcome this challenge.
2. Develop insights into the barriers to funding for female entrepreneurs and define potential solutions to these barriers.
3. Develop a user friendly web app using React where users will have access to features such as:

a. Their profile and the profile of other entrepreneur's
b. An online community forum
c. A rewards web app similar to their competitor *'I Fund Women'*.

3. Further Work Conclusions

There are two significant challenges with creating a crowdfunding web-app for female entrepreneurs in Eastern Africa. Firstly, entrepreneurs in socio-economically disadvantaged regions are often driven by need entrepreneurship rather than opportunity entrepreneurship (Giacomin *et al.*, 2011). Opportunity entrepreneurs prefer to invest in future longer term opportunities while need entrepreneurship is driven by survival instincts. KPESA's platform must encourage opportunity entrepreneurship through mentorship, education and building trust in communities.

Secondly, Africa is the least connected continent with only 22% of the population have access to the internet compared to 80% in Europe (IFC, 2023). In Tanzania, KPESA's home market, up to 37% have access to the internet, with 81% of web pages served accessed by mobile phones (Mwakideu, 2021; Kemp, 2022). However, even those with access to internet have low-bandwidth and limited data therefore the client has emphasised the need for a low data, mobile first web app.

Consequently, the dual requirement of this project is providing a functional product with low data requirements. The app design will be simple as I try to maximise functionality. I hope to complement the software development with further research into the funding opportunity in Eastern Africa, and the competitive landscape that will guide the functionality of the app. By doing so, I hope to develop a prototype that KPESA can then build on in future as a platform as they look to introduce advanced features such as the integration of crowdfunding campaigns. I have included a target schedule below.

Outline	Target Date
Terms of reference finalised	Early/mid-December 2022
Back end skeleton completed	Late December 2022/ early January 2023
Web-app template completed	Early January 2023
Literature review and research overview	Mid-January 2023
First major feature Launched	Late January/ Early February 2023
MVP/protype developed	March 2023

C. DESIGN DOCUMENTATION

C.1 Entity Relationship Diagram and Relational Schema

This section diagrammatically demonstrates the relationships between entities using an entity relationship diagram (ERD) and a relational schema. The ERD is used to design the back end tables by defining the relationships between entities. A double line indicates a 'mandatory' relationship, a single line indicates an optional relationship. N indicates many and 1 indicates a single relationship. Therefore, a 1:N relationship between campaigns indicates there can be many pledges for each campaign but only one campaign for every pledge. The ERD is then mapped into a relational schema to describe how the data is structured into tables.

Entity Relationship Diagram

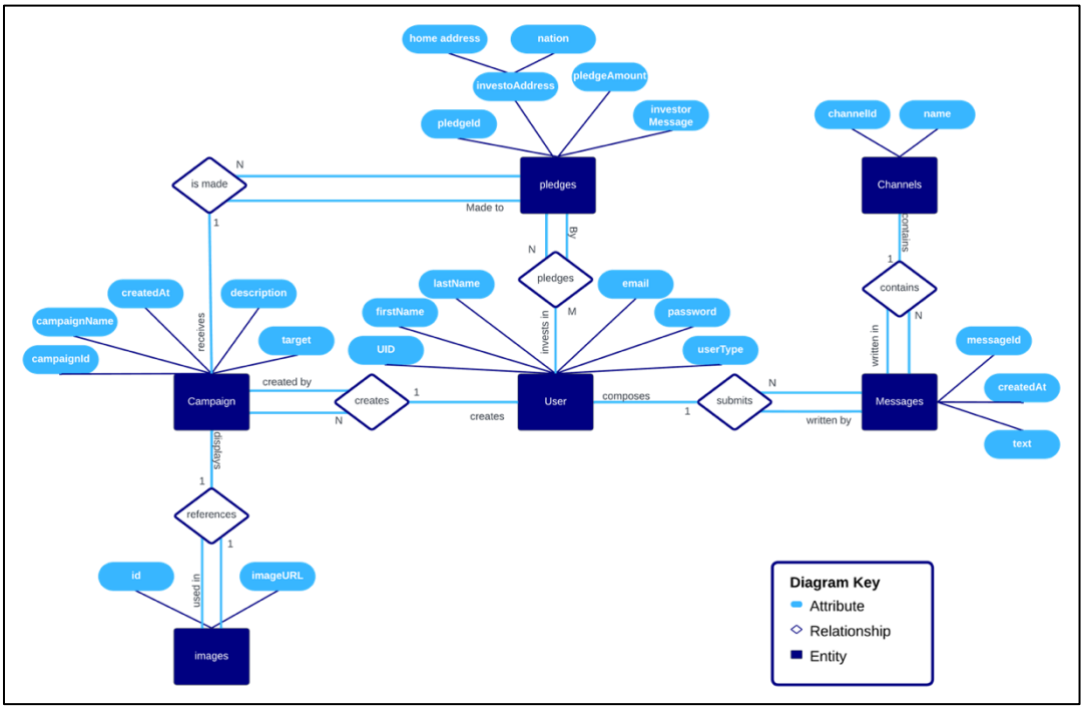


FIGURE C.1.1: Entity relationship diagram for KPesa's web app.

Relational Schema

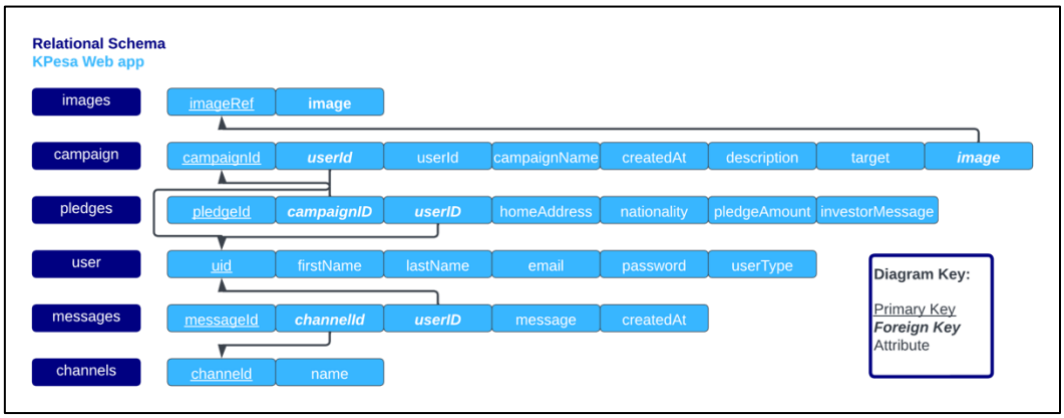


FIGURE C.2.2: A relational schema which was made to ensure consistency in the relationships between entities.

C.2 Data Flow Diagrams

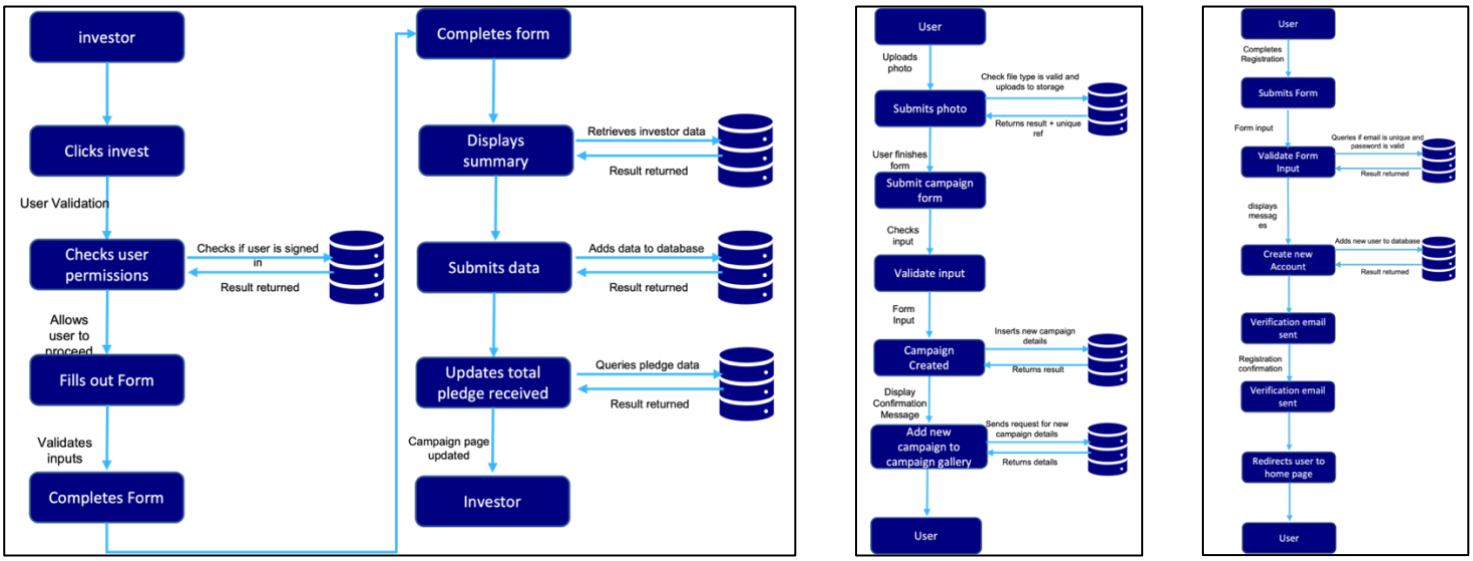


FIGURE C.2.1: Data Flow Diagram for an investment journey (left), building a campaign (middle), and signing up (right)

C.3 Mock-ups

To assist the development of the web app, I built some mock ups in Figma to act as a visual reference. This helped design the composition of elements on the page. Although the design in some instances changed from the mock-up it acted as a useful tool in the design process.

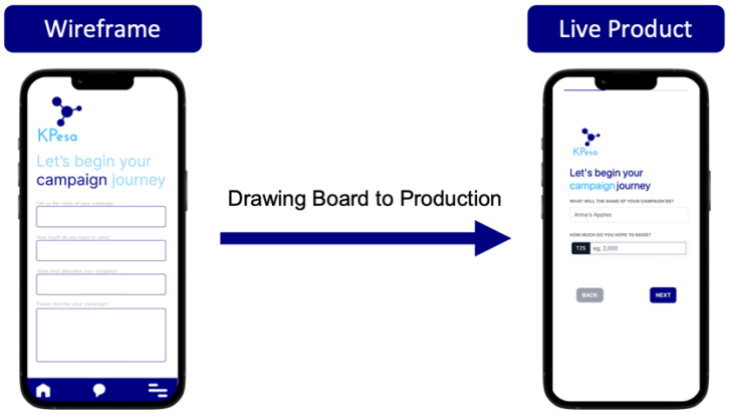


FIGURE C.3.1: An example of the journey from mock-up on Figma to final production

C.4 Software Development Methodology

This section describes the methodology that was used to complete the project. An agile development method was adopted. This involved an iterative approach split into four sections. This was combined with a Kanban board which helped me track the required tasks. The Kanban board was split into a Design, Development, and report phase. Every section would begin with a design requirement. Once the design was completed, I would move onto the

development phase. The completed feature would then be tested using unit testing followed by integration testing. Finally, once the web app was complete it was deployed using Vercel.

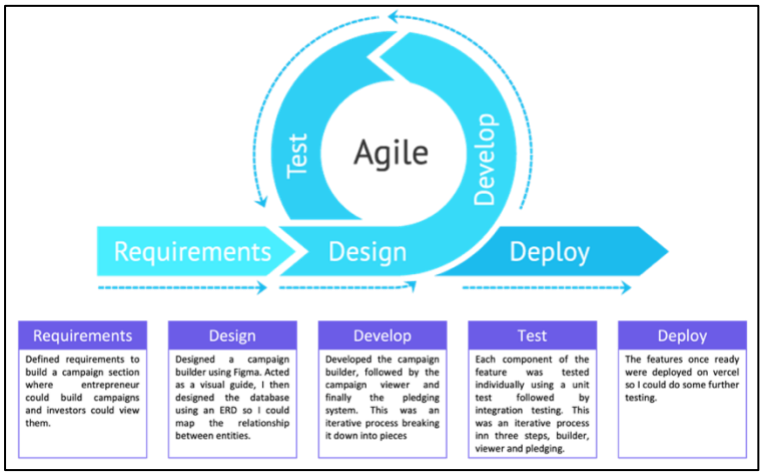


FIGURE C.4.1: Agile Development Image Source: https://www.javatpoint.com/software-engineering-agile-model

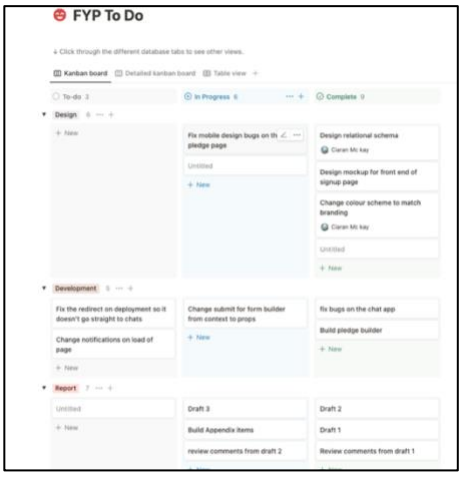


FIGURE C.4.2: A screenshot of the Kanban Board I used to track my progress

C.5 Coding Conventions Used

Coding standards are guidelines on how code should be written. Conventions help developers understand other developer's code. React and JavaScript coding conventions were followed. An opinionated code formatted called Prettier was used to ensure coding conventions were effectively maintained. Some of the coding standards that were used to maintain convention were the following:

- Variable names: Variable names used camel case as an identified. For example, our variable for user data was "userData".
- Indentation: Indentation was used to differentiate between child code blocks and the parent.
- Statements: Every statement finished with a semi-colon.
- Function Names: All function names used camel case. For example a function to retrieve a user's data would be called getUserData().
- Component Names: Component names used a PascalCase naming convention. For example, the component used to track a user's pledge would be called PledgeData.js.
- File names: Mac OS is not case sensitive; however, most web servers are. When deploying the app on Vercel it was important to ensure referencing to files was consistent. One small error in a reference with respect to case could break the website.
- Comments: Comments have been left in the code where explanation was deemed necessary.
- Tags: If a tag did not have a child it would always self-close. For example, an image would always be <Image href ={src}, alt = "sample" />/

D. SAMPLE OF THE APPLICATION SOURCE CODE & INPUT SCREENS

This section provides a link to the code used to build this project and screenshots of some of the pages available to users. Examples are provided for both mobile and desktop experiences.

D.1 Sample of the Application Source Code

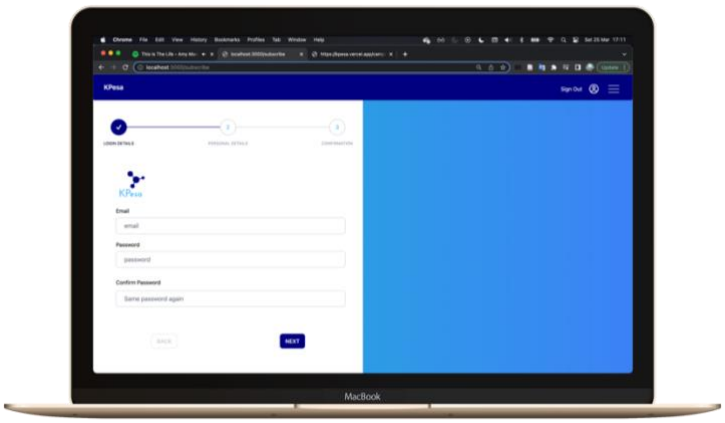
Code for the project can be found in my project GitHub repository.

D.2 Sample Input Screens & Video Demonstration

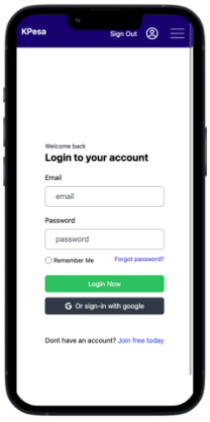
The images below are sample input screens from the web app I have also attached a short ten minute video demonstrating the web app. The video can be found here.

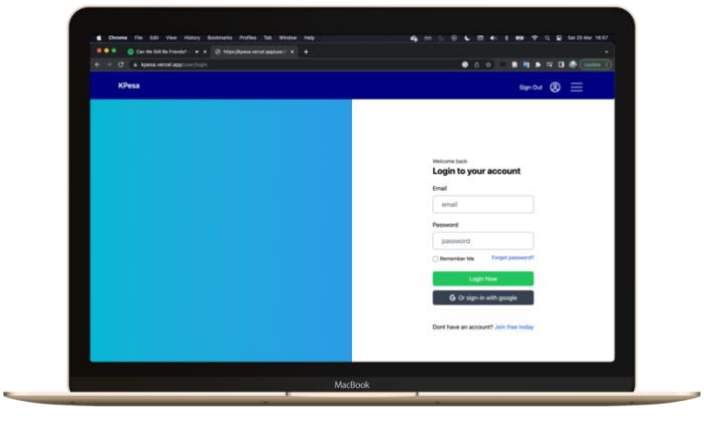
Signup





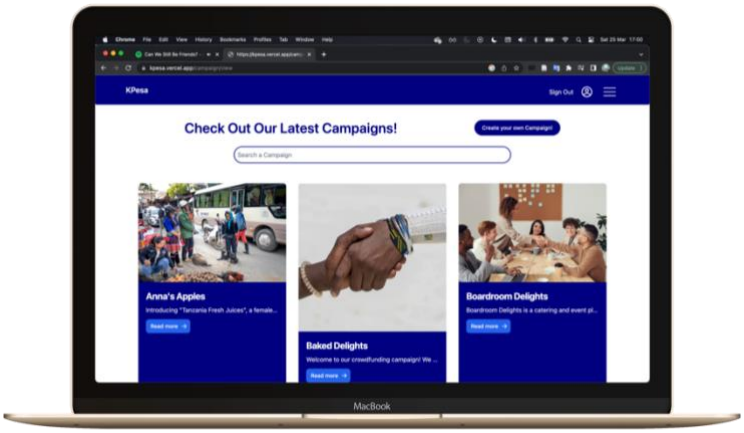
Login



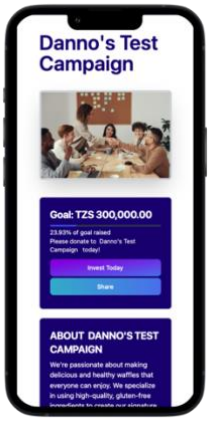


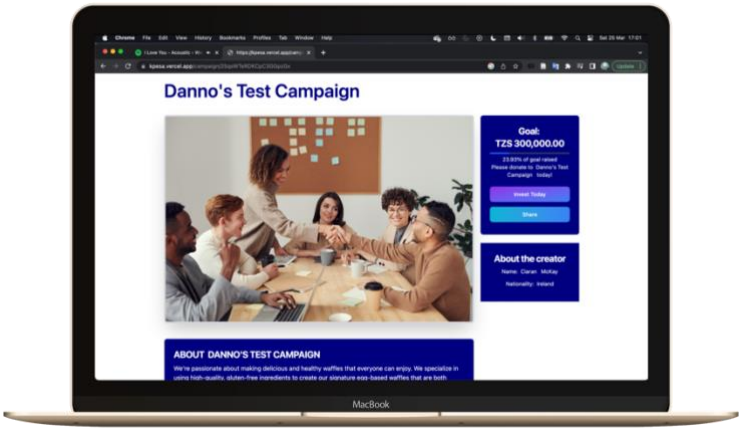
Campaign Gallery





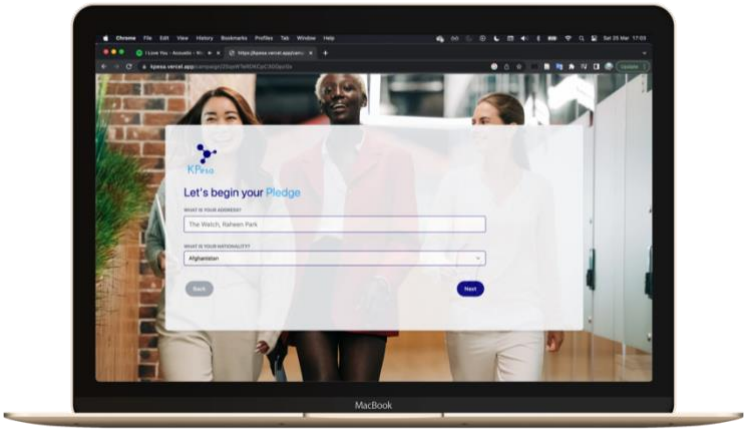
Campaign View





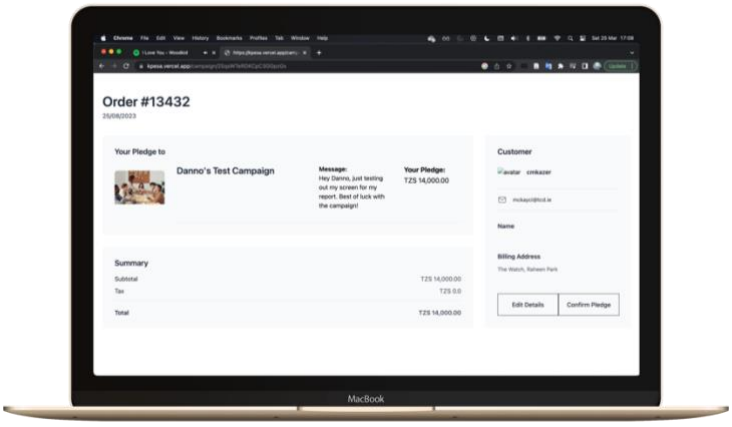
First Step of Campaign Pledge Form



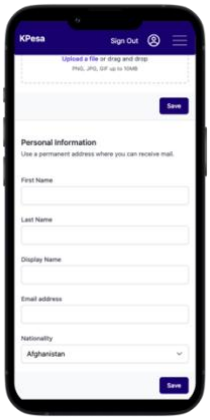


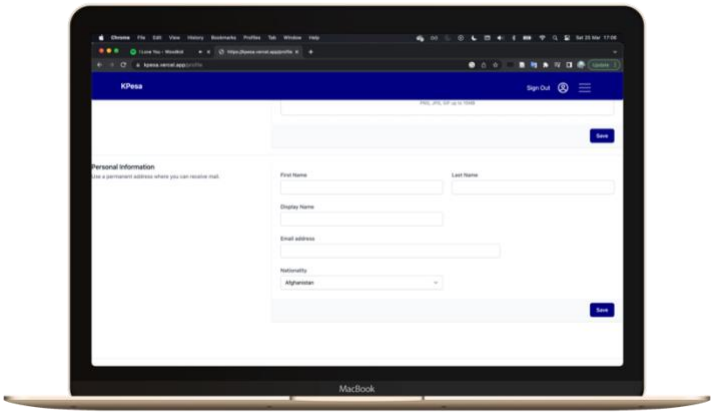
Campaign Pledge Confirmation Page



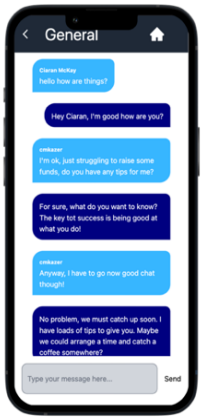


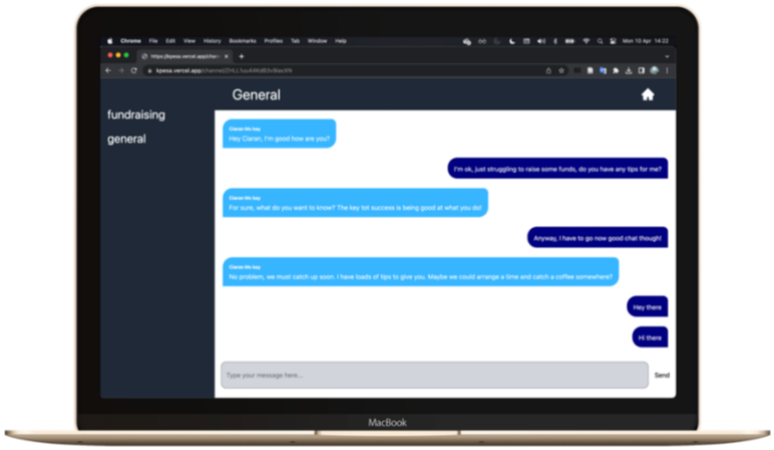
Edit Profile Page





Chat Feature





E. TESTING

This section provides an overview of the testing documentation for the chat app, the campaign section, and the profile section of the web app. The documentation outlines the procedures of testing and the expected results.

E.1 Test Documentation Tables

Table E.1 – Test Documentation for the Authentication Journey

Test	Testing Procedure	Expected Result
Sign-in with Google	Sign in by clicking on the sign in with google form and signing in.	Once signed in the user should be redirected to a password protected page.
Sign-in with Email	Use an authenticated email and password to sign in	The user should be signed in and redirected
Create an account	User should be able to sign up for a new account.	Once signed in they should be redirected and sent an email.
Sign in with invalid information	Sign in with the wrong email	A notification should pop up telling the user their information was incorrect
Sign up with invalid email or non-confirmed password	Sign up with an email that does not contain proper email syntax or try to sign up with a password that is different to confirm password.	User should receive a notification warning them there is a problem with their signup.
Click don't have an account on the login page	Navigate to the login page and click "I don't have an account".	User should be redirected to signup page.
Navigate to protected page	Navigate to a password protected page without being signed in.	Redirected to login page.
logout	Clicks logout in the navbar.	User should be logged out and then redirected to the login page.
Update User details	A user updates their details in the my profile section.	The user's details are updated.

Table E.2 – Test Documentation for Forum

Test	Testing Procedure	Expected Result
Send Message	Type message and send.	The message should appear in the chat.
Navigate to another channel	Click on the name of another channel in the sidebar.	The page should update without a refresh required.
Click back on mobile	User clicks back button on mobile.	Should load the channel navigation pane.
Click hamburger menu	User clicks hamburger menu.	Loads navigation options.
Scroll through messages	Scroll up on phone or desktop to view older messages.	User should see older messages without a refresh.

Table E.3 – Test Documentation for Campaign Section

Test	Testing Procedure	Expected Result
Navigate to campaigns	Click campaigns in hamburger menu.	This should render a campaign gallery.
View campaign	Click on an individual campaign.	Individual campaign details should render.
Navigate to create campaign	User clicks on “create your own campaign button”.	Navigated to create campaign.
Navigate to create campaign when not authenticated	Non-entrepreneur clicks on button.	Redirected to login and a notification appears telling the user why they cannot access that page.
Create campaign	Fill in details and create campaign.	Campaign should be added to database
Upload campaign image	Upload campaign image.	Successful upload should be notified
Upload non-image	Upload a pdf as campaign image.	The file explorer should grey out the submit button
Share campaign	Click on share button and share campaign.	Share should redirect to respective source of share
Invest in campaign	User clicks on invest.	User can invest
Invest if not signed in	Unregistered user clicks invest.	User action blocked, then a notification is displayed informing the user why they have been blocked.

E.2 Vercel Deployment Analytics

The app is deployed on Vercel, which was used to analyse the performance of the app. The images below show the performance of the app based on the speed and smoothness of the app. An overall score above 80 is considered good. When first launched the app received a 59 overall and measures were taken such as changing how images were rendered to improve the performance of the app. How data was handled and stored in the cache upon log in was also changed as the original method was causing inappropriate error messages to display which led to an unpleasant experience and a poor user experience score. The screenshots indicate the score for the 99th percentile of user accounting for 99% of user’s experience. This is an important simulation of performance in areas with slower internet bandwidth.

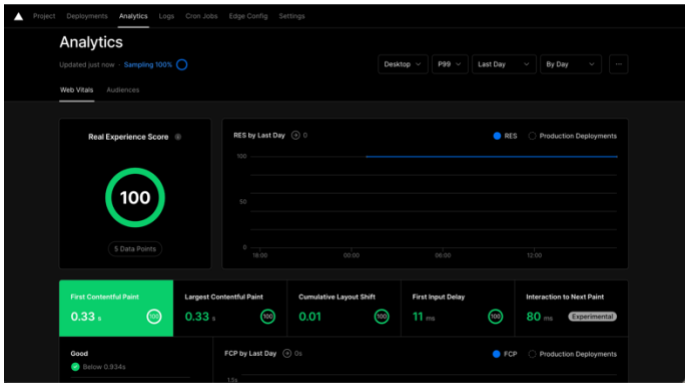


FIGURE E.2.1: A screenshot of the performance for a desktop user

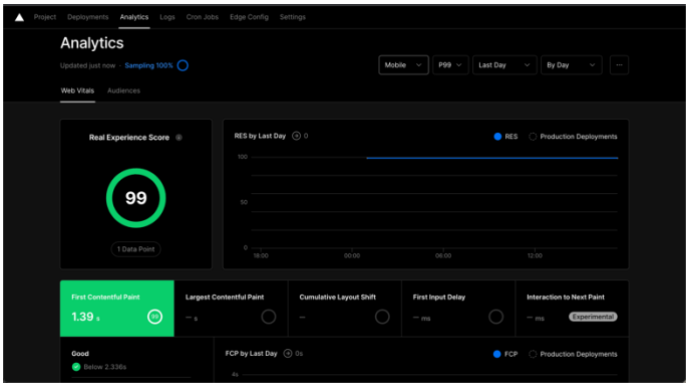


FIGURE E.2.2: A screenshot of the performance for a mobile user

F. ADDITIONAL MATERIAL PRODUCED FOR THE CLIENT

F.1 Firebase Administrator Guide

A guide was created to help the administrator of the database navigate the firebase admin portal easily. This 11-page document covers topics such as Authentication, Storage and editing the database. The first 6 pages of the document can be viewed below. The full guide can be found here.

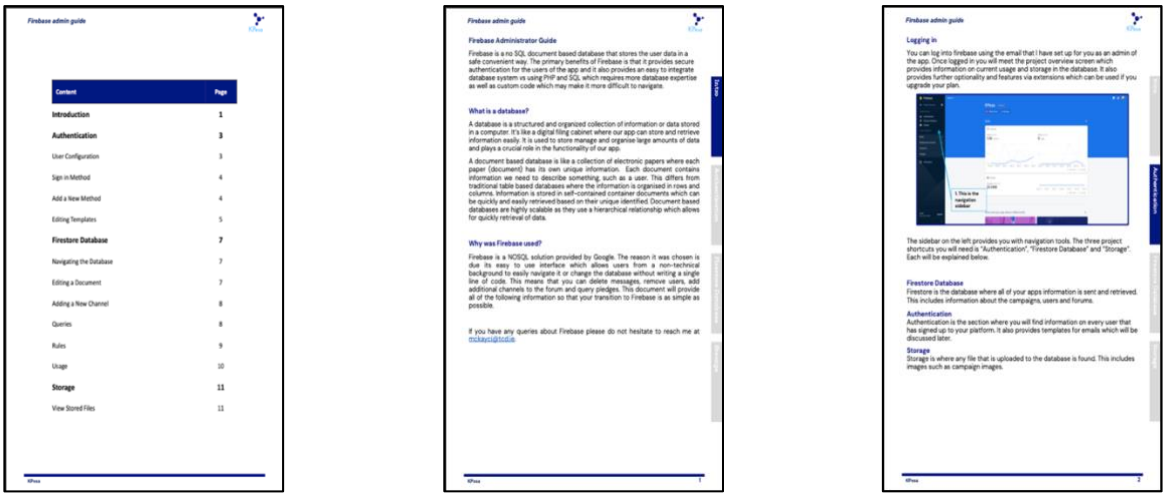


FIGURE F.1.1: Screenshots of the administrator guide provided to KPesa

F.2 Exploration of payment gateways

As part of the scoping, I decided to research the payment gateways used by different Crowdfunding platforms. In total I looked at eight different platforms and included seven candidates which are worth further exploration.

Payment Gateway	Native Crowdfunding Support	Split Payout	Native KYB	Cost	Payment acceptance	Pay out to mobile money	Operates in Tanzania	Compliance	Notable crowdfunding customers
MangoPay	✓	✓	✓	1.18% + €0.18 per transaction KYB: €2.50/entity	Apple Pay, Google Pay, PayPal, GiroPay, iDeal, Payconiq	✗	✓	E-money licenses in EU PSD2 SCA compliant PCI DSS Level 1	raizers, LITA.co
Tingg	✗	✗	✗	Domestic: 1.5% International: 3.2%	Cards, Bank Transfer, Mobile Money	✓	✓	PCI DSS Level 1	
LemonWay	✓	✓	✓	International:2.8% + €0.18 SEPA: Free	CB, Mastercard, Visa, SEPA, Cheques	✗	?	PCI DSS standards RGPD standards certified	october, ESTATEGURU
GoCardless	✗	✓	✗	EU: 1% + €0.20 Else: 2% + €0.20 Different plans offered	Direct Debit	✗	✗	ISO27001 certified for information security management	InvestMySchool
FlutterWave	✗	✓	✗	Tanzania:3.8% transfer to bank account: €1.18 Mobile money <€8: €0.16 Mobile Money >€8: 3%	Cards, Mpesa, Bank Transfer, Mobile Money	✓	✓	ISO 27001 & 22301 Certification	
Stripe Connect	✓	✓	✓	€2 per MAU Payout: 0.25% + €0.32 KYB: €1.99	Cards, Bank Debits, Transfers, BNPL, Vouchers, Google Pay, Apple Pay	✗	✓	PCI DSS Level 1	KICKSTARTER, crowdcube
PayPal	✗	✓	✓	+3.5% + fixed fee 4.0% FX	PayPal, Credit Card, Bank Transfer	✗	✓	PCI DSS Level 1 and 3DS 2.0	kiva, DonorDrive

FIGURE F.2.1: A table detailing the different features of major payment gateways

F.3 Crowdfunding Business Models

Donation Based Model

Donation based crowdfunding is the most popular method of crowdfunding in Africa. Major donation platforms for entrepreneurs include KIVA which operates as a charity. There is no finance return associated with donated based crowdfunding models.

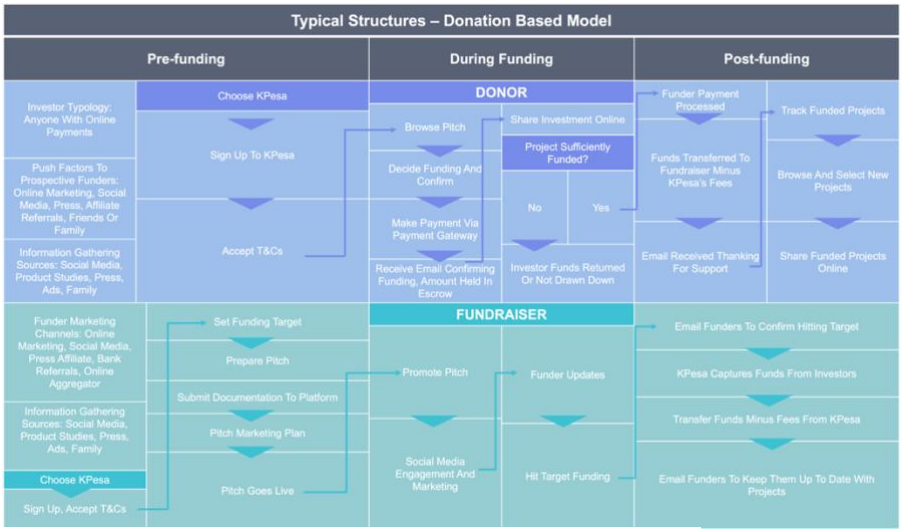


FIGURE F.3.1 Source: (Garvey et al., 2017)

Equity Based Model

Equity crowdfunding is the second most popular crowdfunding model, accounting for 10% of crowdfunding proceeds globally (Garvey *et al.*, 2017). Regulations is emphasised due to the typically higher risk profile of equity crowdfunding.

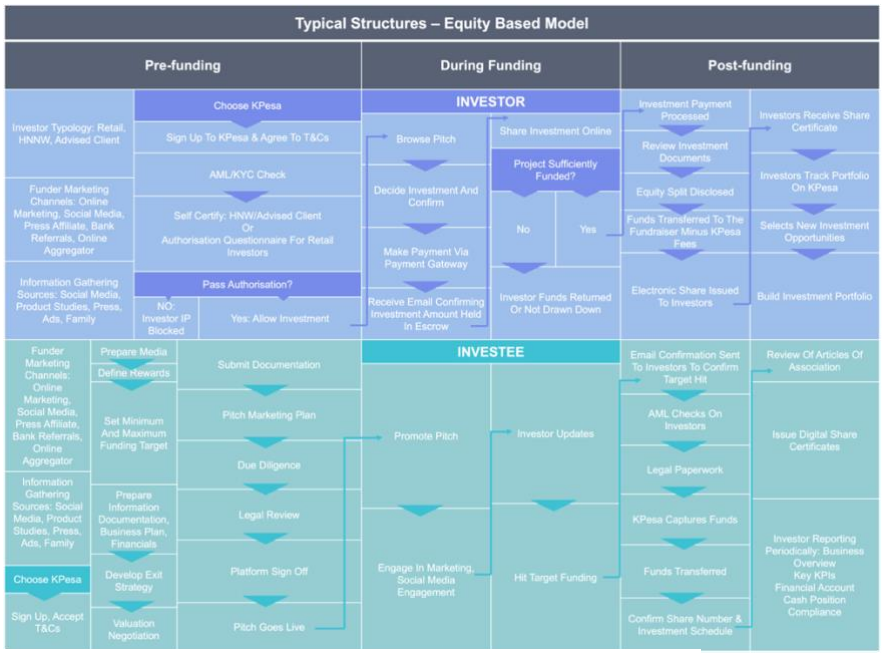


FIGURE F.3.2 Source: (Garvey et al., 2017)

F.4 Survey Of Female Entrepreneurs In Tanzania

Motivations of Entrepreneurship

Entrepreneurial Motivations (top 5)	Mean (/5)
It was the only way I could make a living	4.58
I could not rely on my spouse's income	4.48
To increase my income	4.45
Unable to find a paid job	4.20
To be able to do the kind of work I want to do	4.02

Entrepreneurial Motivations (bottom 5)	Mean (/5)
To follow the example of a person I admire	2.41
To continue family tradition	2.38
My salary was not enough to sustain my family	2.17
Experienced Discrimination in my workplace	1.82
To due redundancy	1.74

Challenges Facing Female Entrepreneurs

Challenges Faced (top 5)	Mean (/5)
Unable to obtain short-term financial capital	4.58
Unable to obtain long-term financial capital	4.48
Lack of property rights over assets for collateral	4.46
Good premises and working tools	4.41
Lack of confidence in women by bank officers	4.20

Challenges Faced (bottom 5)	Mean (/5)
Lack of marketing training	2.48
Complex/confusing tax structure	2.38
Complicated business registration process	2.17
Poor roads/transportation	1.2
Electricity problems	1.75

Firm Profile

Number of Employees	%
1 – 4 employees (micro)	68.75
5 – 49 employees (small)	31.25
50 – 99 employees (medium	0

Industry	%
Service	31.25
Food Processing	30
Tailoring mart/garment making	38.75

Age Of The Firm	%
5 years	13.75
6-10 years	41.25
10-15 years	32.5
16-20 years	9.75
21-25 years	2
Over 26 years	0.5

Legal Status	%
Sole Proprietorship	80
Partnership	20

All FIGURES: Source: (Isaga, 2018)

G. NEXT STEPS AND CLIENT FEEDBACK

G.1 Next Steps

KPesa will be onboarding three companies seeking investment and have contacted several interest investors. They plan to use the web app to run a beta test with the focus group and gather feedback based on empirical user data. The live beta test will help inform KPesa on additional features and bug fixes that are required.

To ensure a smooth transition, I have offered to spend some additional time onboarding beta testers in the early Summer, modifying the web app and fix additional bugs found based on feedback from users. I hope to offer my services in an informal capacity on a voluntary basis to ensure the success of this important cause.

G.2 Client Feedback

Following the conclusion of the final meeting of this project, the client provided the following feedback.

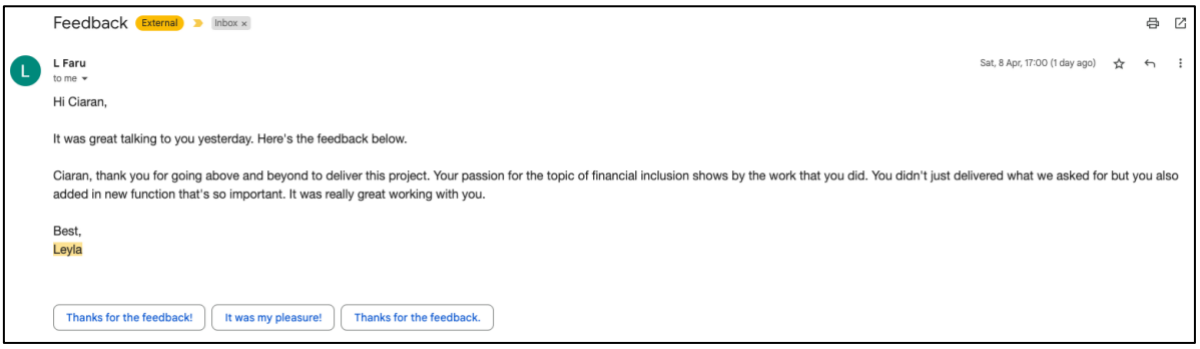


Figure G.2.1: A screenshot of an email sent from the client providing some brief feedback

H. REFERENCES

Akinbami, C. and Aransiola, J. (2015) 'Qualitative exploration of cultural practices inhibiting rural women entrepreneurship development in selected communities in Nigeria', *Journal of Small Business & Entrepreneurship*, 28, pp. 1–17. Available at: https://doi.org/10.1080/08276331.2015.1102476.

Amon, S., Ojong, N. and Dana, L.-P. (2021) *Female entrepreneurship in Africa: A review, trends, and future research directions - ScienceDirect*. Available at: https://www.sciencedirect.com/science/article/pii/S0148296321002708 (Accessed: 29 March 2023).

Bardasi, E., Sabarwal, S. and Terrell, K. (2011) 'How do female entrepreneurs perform? Evidence from three developing regions', *Small Business Economics*, 37(4), pp. 417–441.

Brüderl, J. and Preisendörfer, P. (1998) 'Network Support and the Success of Newly Founded Businesses', *Small Business Economics*, 10(3), pp. 213–225.

CRDB Bank (2023) *CRDB Bank PLC | Loan Calculator*. Available at: https://crdbbank.co.tz/en/loan-calculator/business/21 (Accessed: 29 March 2023).

CSMA (2022) *News |Capital Market and Securities Authority*. Available at: https://www.cmsa.go.tz/news/request-for-comments-on-draft-capital-markets-and-securities-crowdfunding-guidelines-2022 (Accessed: 23 March 2023).

DatabaseTown (2023) 'Document Database VS Relational Database - DatabaseTown', 21 January. Available at: https://databasetown.com/document-database-vs-relational-database/ (Accessed: 24 February 2023).

Dushime, A. (2022) *How female led startups in Africa can uplift the continent*, *World Economic Forum*. Available at: https://www.weforum.org/agenda/2022/08/how-female-led-start-ups-can-transform-africa/ (Accessed: 2 December 2022).

EU (2023) *Crowdfunding explained*. Available at: https://single-market-economy.ec.europa.eu/access-finance/guide-crowdfunding/what-crowdfunding/crowdfunding-explained_en (Accessed: 29 March 2023).

Garvey, K. *et al.* (2017) 'Crowdfunding in East Africa: Regulation and Policy for Market Development'. Rochester, NY. Available at: https://doi.org/10.2139/ssrn.3621303.

Giacomin, O. *et al.* (2011) *Opportunity and/or necessity entrepreneurship? The impact of the socio-economic characteristics of entrepreneurs*. Available at: https://mpra.ub.uni-muenchen.de/29506/ (Accessed: 2 December 2022).

Halberstadt, J. and Spiegler, A. (2018) 'Networks and the idea-fruition process of female social entrepreneurs in South Africa', *Social Enterprise Journal*, 14. Available at: https://doi.org/10.1108/SEJ-01-2018-0012.

Herbert, D. (2022) *What is React.js? (Uses, Examples, & More)*. Available at: https://blog.hubspot.com/website/react-js (Accessed: 2 February 2023).

IFC, . (2023) *Bringing Africa Up to High Speed*. Available at: https://www.ifc.org/wps/wcm/connect/NEWS_EXT_CONTENT/IFC_External_Corporate_Site/News+and+Events/News/CM-Stories/CM-Connecting-Africa (Accessed: 2 December 2022).

Inscribe (2023) *Digital Account Opening Solutions | Inscribe*. Available at: https://www.inscribe.ai/account-opening (Accessed: 27 February 2023).

IOS App Store (2023) *discord - Apple (IE)*. Available at: https://www.apple.com/ie/search/discord?src=globalnav (Accessed: 27 March 2023).

Isaga, N. (2018) 'Start-up motives and challenges facing female entrepreneurs in Tanzania', *International Journal of Gender and Entrepreneurship*, 11(2), pp. 102–119. Available at: https://doi.org/10.1108/IJGE-02-2018-0010.

Jason Miller and Osmani, A. (2022) *Rendering on the Web*, *web.dev*. Available at: https://web.dev/rendering-on-the-web/ (Accessed: 21 March 2023).

Kemp, S. (2022) 'DIGITAL 2022: TANZANIA', *DataReportal*, 15 February. Available at: https://datareportal.com/reports/digital-2022-tanzania#:~:text=Internet%20use%20in%20Tanzania%20in,at%20the%20start%20of%202022.

McAdam, M. (2022) *Women's Entrepreneurship*. Taylor & Francis.

Mersha, T. and Sriram, V. (2019) 'Gender, entrepreneurial characteristics, and success: Evidence from Ethiopia', *Thunderbird International Business Review*, 61(2), pp. 157–167. Available at: https://doi.org/10.1002/tie.21984.

Mwakideu, C. (2021) 'Can Africa Reach Universal Internet Access?', *DW*, 11 May. Available at: https://www.dw.com/en/can-africa-achieve-universal-internet-access-by-2030/a-59729090.

Ngowi, H.P. (2021) 'Use of Whatsapp Social Media as Platform for Crowdfunding: Selected Observations in Tanzania', *Eastern Africa Social Science Research Review*, 37(2), pp. 43–56. Available at: https://doi.org/10.1353/eas.2021.0008.

Panda, S. (2018) 'Constraints faced by women entrepreneurs in developing countries: review and ranking', *Gender in Management: An International Journal*, 33. Available at: https://doi.org/10.1108/GM-01-2017-0003.

Pittet, S. (2023) *The different types of testing in software*, *Atlassian*. Available at: https://www.atlassian.com/continuous-delivery/software-testing/types-of-software-testing (Accessed: 27 February 2023).

Rutashobya, L.K. (2001) 'Female Entrepreneurship in Tanzania: Constraints and Strategic Considerations'. Available at: http://repository.businessinsightz.org/handle/20.500.12018/197 (Accessed: 30 March 2023).

Vercel (2023) *Learn | Next.js*. Available at: https://nextjs.org/learn (Accessed: 2 February 2023).

Witbooi, M. and Ukpere, W. (2011) 'Academic Journals Indigenous female entrepreneurship: Analytical study on access to finance for women entrepreneurs in South Africa', *African journal of business management*, 5, pp. 5646–5657. Available at: https://doi.org/10.5897/AJBM10.1161.

World Bank (2021) *Fixed broadband subscriptions (per 100 people) - Tanzania | Data*. Available at: https://data.worldbank.org/indicator/IT.NET.BBND.P2?locations=TZ (Accessed: 2 February 2023).

Zapier (2023) *Connect your Flutterwave to Stripe integration in 2 minutes*, *Zapier*. Available at: https://zapier.com/apps/flutterwave/integrations/stripe (Accessed: 23 March 2023).